

A Project Report On

A Statistical Analysis of Systematic Investment Plans (SIP) in Mutual Funds

Submitted in partial fulfillment of the requirement for the 4th semester

Master of Science

in

STATISTICS

DEPARTMENT OF STATISTICS,

SCHOOL OF MATHEMATICAL SCIENCES,

KAVAYITRI BAHINABAI CHAUDHARI NORTH MAHARASHTRA
UNIVERSITY



Submitted By

Mr.Badgujar Krushna Padmakar (358771)

Ms.Pawar Sapana Anil (358709)

Ms.Sonawane Kalyani Anil (358799)

Under the guidance of

Dr. R. D. Koshti

Department of Statistics , K.B.C N.M.U.

(2024-2025)

**KAVAYITRI BAHINABAI CHAUDHARI NORTH MAHARASHTRA
UNIVERSITY**

SCHOOL OF MATHEMATICAL SCIENCES

DEPARTMENT OF STATISTICS



CERTIFICATE

This is to certify that the project entitled **A Statistical Analysis of Systematic Investment Plans (SIP) in Mutual Funds** is a bonafide work carried out by **BADGUJAR KRUSHNA PADMAKAR, PAWAR SAPANA ANIL, SONAWANE KALYANI ANIL** in partial fulfillment of 4th semester, Master of Science in STATISTICS under Kavayitri Bahinabai Chaudhari University North Maharashtra University, Jalgaon during the year 2024-25.

Dr.R.D.Koshti

(Internal Guide)

Department of Statistics, KBCNMU

Prof.K.K.Kamalja

HOD

Dept. of Statistics, KBCNMU

Acknowledgement

We are pleased to have successfully completed the project **A Statistical Analysis of Systematic Investment Plans (SIP) in mutual funds**. I thoroughly enjoyed the process of working on this project and gained a lot of knowledge doing so.

we would like to take this opportunity to express our gratitude to **Prof. (Mrs.) K. K. Kamalja**, Head of Department, Department of Statistics, for permitting me to utilize all the necessary facilities of the institution.

We are immensely grateful to my respected and learned guide, **Dr. R. D. Koshti**, Assistant Professor, Department of Statistics for his valuable help and guidance. we are indebted to him for his invaluable guidance throughout the process and his useful inputs at all stages of the process.

We also thank all the faculty and support staff of the Department of Statistics of the School of Mathematical Sciences. Without their support, this work would not have been possible.

Lastly, We would like to express our deep appreciation towards our classmates and our family for providing us with constant moral support and encouragement. They have stood by us in the most difficult of times.

Mr.Krushna Badgujar (358771)

Ms.Sapana Pawar (358709)

Ms.Kalyani Sonawane (358799)

Abstract

This study investigates investor preferences in Systematic Investment Plans (SIPs), focusing on factors such as expected return, annual income, occupation, and investment amount. As SIPs gain popularity for structured and long-term wealth building, understanding the behavioral patterns of investors becomes vital.

The objective of this project is to analyze the relationship between demographic and financial variables and the choice of SIP type (Equity or Debt) as well as return expectations.

The dataset was sourced from the Groww app and analyzed using descriptive statistics and chi-square tests of independence to determine significant associations.

Findings reveal that income level has a significant relationship with SIP preference—higher income groups tend to prefer debt plans more than lower income groups. Government employees show the highest inclination towards equity plans. Additionally, higher SIP amounts are associated with higher return expectations.

The results emphasize the importance of aligning investment advice with investor profiles, particularly income and return expectations, to optimize satisfaction and financial outcomes.

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Chapter 1

Introduction

1.1 Background

Stock Market is considered as one of the major indicators of an economy. The stock market plays a pivotal role in any nation's economy by providing a platform for companies to raise capital and for investors to grow their wealth. It serves as a barometer for economic performance, reflecting market sentiment, economic growth, and industrial activity. The Indian stock market, one of the largest and fastest-growing in the world, holds a significant place in the global financial ecosystem.

India's stock market ecosystem is primarily composed of two leading exchanges: the Bombay Stock Exchange (BSE) and the National Stock Exchange (NSE). The BSE, established in 1875, is the oldest stock exchange in Asia and the cornerstone of India's financial sector. The NSE, established in 1992, introduced electronic trading in India, revolutionizing the trading process. These exchanges together form the backbone of India's financial market.

Among the indices tracked by the BSE, the Sensex, short for the Sensitive Index, is the benchmark index. It comprises 30 of the largest and most actively traded companies across diverse sectors, providing a snapshot of market performance. Since its inception in 1986, the Sensex has become a reliable indicator of economic and market trends, reflecting the overall health of the Indian economy.

1.2 Objectives

- Evaluate the performance of selected top mutual funds through Systematic Investment Plans (SIPs).

- Compare the performance of mutual fund schemes through SIPs.
- Analyze trends in NAV over time using time series analysis.
- Summarize returns with descriptive statistics (CAGR, XIRR, standard deviation).
- Apply PCA to reduce dimensionality and identify key performance features.
- Use K-Means Clustering on PCA features to group funds into performance-based clusters.

1.3 Mutual Funds

A mutual fund is a pool of money managed by a professional Fund Manager. It is a trust that collects money from a number of investors who share a common investment objective and invests the same in equities, bonds, money market instruments and/or other securities. And the income / gains generated from this collective investment is distributed proportionately amongst the investors after deducting applicable expenses and levies, by calculating a scheme's "Net Asset Value" or NAV. Simply put, the money pooled in by a large number of investors is what makes up a Mutual Fund.

1.3.1 Types of Mutual Funds

Mutual funds are typically categorized based on the kind of assets they invest in, and their objective. The main types include: Investors should choose funds that align with their risk tolerance and investment goals. Equity funds are generally suitable for those with a higher risk appetite and a long-term investment horizon, while debt funds may be more appropriate for conservative investors seeking stable returns. Hybrid funds offer a middle ground by balancing risk and return through diversified investments in both asset classes.

- **Equity Funds:** Invest primarily in stocks. These funds are designed for long-term capital growth and carry higher risk.
- **Debt Funds:** Invest in fixed-income securities like bonds. These are lower risk and generate regular income but have lower returns compared to equity funds.
- **Hybrid Funds:** Invest in a mix of equities and bonds. These funds aim to offer a balance between risk and return.
- **Index Funds:** Track a specific index, like the Nifty 50 or Sensex, and replicate its performance. These are passive funds and usually have lower fees.
- **Sectoral Funds:** Focus on specific sectors of the economy (e.g., technology, health-care). They tend to be more volatile but can offer high returns.

1.3.2 Benefits of Mutual Funds

- **Diversification:** Mutual funds pool money from many investors and invest it in a range of securities, offering diversification and reducing individual risk.
- **Professional Management:** Mutual funds are managed by experienced professionals who make investment decisions based on research and analysis.
- **Liquidity:** Most mutual funds allow you to redeem your investment easily, though there may be some lock-in periods for specific funds like ELSS (Equity Linked Savings Scheme).
- **Affordability:** With a small amount of money, you can get exposure to a wide range of securities, which would be difficult to achieve through direct investing in stocks or bonds.

1.3.3 Basic Mutual Fund Terms

AUM (Assets Under Management): AUM refers to the market value of the total fund or the assets the fund holds. It is an indicator of the success of the fund. The bigger the AUM, the higher the trading value. This simply indicates that a large number of investors have invested in the corpus already and what kind of operations they would be dealing with. It also gives a bit of an idea of what the fees might be like for a certain fund.

CAGR(Compound Annual Growth Rate) : CAGR is the average annual growth rate of an investment over a specified period, assuming the investment grows at a steady rate compounded annually.

$$\text{CAGR} = \left(\frac{\text{Ending Value}}{\text{Beginning Value}} \right)^{\frac{1}{n}} - 1$$

XIRR (extended internal rate of return) XIRR stands for Extended Internal Rate of Return is a method used to calculate returns on investments where there are multiple transactions happening at different times.

In the case of SIP, since there are multiple investments (therefore multiple purchase prices) and different time periods for each installment, calculating returns is a bit tricky. Returns on mutual fund SIP is commonly done in terms of XIRR.

Standard Deviation: To measure the volatility or risk correlated with investment. Standard deviation helps to evaluate the volatility in returns of the average return.

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (R_i - \bar{R})^2}{N}}$$

Where:

R_i = Return in period i ,

$\bar{R}$ = Average Return,

N = Number of observations.

Expense Ratio: The annual fee expressed as a percentage of a fund's assets. A lower expense ratio typically means higher net returns over time.

$$\text{Expense Ratio} = \left(\frac{\text{Total Fund Expenses}}{\text{Average AUM}} \right) \times 100$$

1.4 Classification of Mutual Funds

1.4.1 Overview

Mutual funds can be broadly categorized based on the type of securities they invest in, the investment objectives, and the risk-return profile. This classification helps investors choose funds aligned with their financial goals and risk appetite.

1.4.2 Debt Funds

Debt funds primarily invest in fixed-income securities such as government bonds, corporate bonds, treasury bills, and money market instruments.

Low Duration Funds

- **Definition:** Invest in instruments with a Macaulay duration between 6 to 12 months.
- **Risk Level:** Low to Moderate
- **Returns:** Higher than liquid and ultra-short funds
- **Best For:** Short-term investors seeking better returns than savings accounts

Medium Duration Funds

- **Definition:** Invest in instruments with a duration of 3 to 4 years.
- **Risk Level:** Moderate
- **Returns:** Better than short-term debt funds
- **Best For:** Medium-term investors

Dynamic Bond Funds

- **Definition:** Actively manage portfolio based on interest rate movements.
- **Risk Level:** Moderate to High
- **Returns:** Market-linked
- **Best For:** Investors seeking flexible debt allocation

Gilt Funds

- **Definition:** Invest in government securities
- **Risk Level:** Low credit risk, high interest rate risk
- **Returns:** Depend on interest rate changes
- **Best For:** Safety-oriented investors

Credit Risk Funds

- **Definition:** Invest in lower-rated corporate bonds
- **Risk Level:** High
- **Returns:** Higher than other debt funds
- **Best For:** Investors with high risk tolerance

Liquid Funds

- **Definition:** Invest in very short-term instruments with high liquidity
- **Risk Level:** Very Low
- **Returns:** Lower but stable
- **Best For:** Parking idle cash

Ultra Short Duration Funds

- **Definition:** Invest in instruments with a duration of 3 to 6 months
- **Risk Level:** Low
- **Returns:** Slightly better than liquid funds

- **Best For:** Short-term needs

1.4.3 Hybrid Funds

Hybrid funds invest in a mix of equity and debt instruments.

Aggressive Hybrid Funds

- **Equity Exposure:** 65–80%
- **Risk Level:** Moderate to High
- **Returns:** Higher than conservative hybrid funds
- **Best For:** Long-term growth with some stability
- **Examples:** ICICI Prudential Equity & Debt Fund, HDFC Hybrid Equity Fund

Conservative Hybrid Funds

- **Equity Exposure:** 10–25%
- **Risk Level:** Low to Moderate
- **Returns:** Higher than debt funds
- **Best For:** Investors looking for stable income
- **Examples:** HDFC Hybrid Debt Fund, SBI Conservative Hybrid Fund

Arbitrage Funds

- **Strategy:** Exploit price differences in spot and futures markets
- **Risk Level:** Very Low
- **Returns:** Similar to liquid funds
- **Best For:** Low-risk investors seeking tax efficiency

1.4.4 Equity Funds

Equity mutual funds invest primarily in stocks and are classified based on company size or investment strategy.

Multi Cap Funds

- **Definition:** Invest in large, mid, and small-cap stocks

- **Risk Level:** Moderate to High
- **Returns:** High
- **Examples:** Kotak Multi Cap Fund, SBI Multi Cap Fund

Large Cap Funds

- **Definition:** Invest minimum 80% in top 100 large-cap companies
- **Risk Level:** Low to Moderate
- **Returns:** Consistent
- **Examples:** HDFC Top 100 Fund, ICICI Prudential Bluechip Fund

Mid Cap Funds

- **Definition:** Invest at least 65% in mid-cap companies (101st–250th)
- **Risk Level:** High
- **Returns:** High
- **Examples:** Axis Midcap Fund, Mirae Asset Midcap Fund

Small Cap Funds

- **Definition:** Invest at least 65% in small-cap companies (251st and below)
- **Risk Level:** Very High
- **Returns:** High (long-term)
- **Examples:** SBI Small Cap Fund, Nippon India Small Cap Fund

ELSS (Equity Linked Savings Scheme)

- **Definition:** Tax-saving fund under Section 80C with 3-year lock-in
- **Risk Level:** Moderate to High
- **Returns:** High
- **Examples:** Axis Long Term Equity Fund, Mirae Asset Tax Saver Fund

Sectoral/Thematic Funds

- **Definition:** Invest in specific sectors (e.g., IT, Pharma, Banking)
- **Risk Level:** Very High
- **Returns:** Can be high but sector-dependent
- **Examples:** SBI Banking & Financial Services Fund, Nippon India Pharma Fund

1.4.5 Summary Table

Category	Type	Duration	Risk	Returns	Ideal For
Debt	Low Duration	6–12 months	Low to Moderate	Stable	Short-term savers
	Medium Duration	3–4 years	Moderate	Moderate	Medium-term investors
	Credit Risk	3–5 years	High	High	High-risk investors
	Gilt	Flexible	Low Credit, High IR Risk	Variable	Risk-averse investors
Hybrid	Aggressive	Long-Term	Moderate to High	High	Balanced growth
	Conservative	Short to Mid	Low to Moderate	Moderate	Safety seekers
	Arbitrage	Short-Term	Very Low	Stable	Tax-efficient short-term
Equity	Large Cap	Long-Term	Low to Moderate	Moderate	Conservative equity
	Mid Cap	Long-Term	High	High	Risk-tolerant investors
	Small Cap	Long-Term	Very High	Very High	Long-term aggressive
	Multi Cap	Long-Term	Moderate to High	High	Diversified investors
	ELSS	3 Years Lock-in	Moderate to High	High	Tax-saving investors
	Sectoral/Thematic	Depends	Very High	Variable	Sector-specific

Table 1.1: Summary of Mutual Fund Types

Chapter 2

Literature Review

Year	Title	Author(s)		Description
2020	<i>“The Comparative study on Systematic Investment Plan and One Time Investment Plan in Mutual Fund”</i>	Sahana Choudhari	R	The research paper compares SIP and lump-sum investment to help investors choose wisely. CAGR is used for lump-sum plans, and XIRR for SIPs. It concludes that one-time investment is preferable for those with a lump sum.
2023	<i>“A Study of Investment Style Timing of Mutual Funds in India”</i>	S Pavithra, Kayal	P	The study investigates market timing abilities of Indian mutual fund managers (2014–2019) across small-cap, mid-cap, large-cap, and multi-cap funds using a 4-factor model and Treynor-Mazuy and Henriksson-Merton timing models.
2023	<i>“Performance Evaluation of SIP in Mutual Funds: A Comparative Analysis”</i>	Ms. Komal Bansidhar Sharma		The study evaluates SIP performance in India's top 30 mutual funds. It uses benchmarks like BSE Sensex and 91-day T-bills, and metrics like alpha, beta, Sharpe, Treynor, Sortino ratios, and standard deviation. Results highlight SIP as a useful investment strategy.

Table 2.1: Summary of Literature on SIP and Mutual Fund Timing

Chapter 3

Systematic Investment Plan (SIP)

3.1 What is SIP?

A Systematic Investment Plan (SIP) is a method of investing in mutual funds where investors contribute a fixed amount of money at regular intervals, typically monthly, quarterly, or weekly, rather than making a one-time lump sum investment.

This approach allows for disciplined investing and helps mitigate the impact of market volatility through a strategy known as **rupee-cost averaging**. By investing consistently over time, investors buy more units when prices are low and fewer units when prices are high, which can lead to a lower average cost per unit in the long run.

3.2 Benefits of SIPs

- **Affordability:** SIPs enable investors to start with small amounts, making them accessible even for those with limited incomes.
- **Discipline:** Regular investments foster a habit of saving and investing, which is crucial for long-term financial planning.
- **Flexibility:** Investors can adjust their contributions, pause investments, or switch funds as needed.

3.3 One Time Investment (Lump Sum)

Lump sum investment refers to Investing a considerable amount in a Mutual Fund scheme. In lump sum investing, the money is deposited as a one-time down payment. Individuals who are having a large amount can deposit a significant proportion in Mutual Fund. Lump sum investments are generally made for longer terms which improves the chances of earning higher capital gains. Usually lump sum investments are ventured by big players and investors, in stocks especially those linked to assets that are likely to acknowledge in the long term, making the investment beneficial except in cases of high volatility

3.4 Benefits of SIP Over Lump Sum Investment

1. **Rupee Cost Averaging:** You buy more units when prices are low and fewer when prices are high, lowering your average cost per unit.
2. **Reduced Market Timing Risk:** SIPs eliminate the need to time the market.
3. **Financial Discipline:** Automated investments ensure consistent contributions toward goals.
4. **Flexibility:** Start small (even Rs 100), pause or modify investments.
5. **Compounding Benefits:** Long-term SIPs maximize compounding power.
6. **Lower Risk for Beginners:** Especially suited for those with lower risk tolerance.

3.5 How Long Can You Hold a SIP?

There is no fixed limit to the duration for which you can continue a SIP. You can hold SIPs as long as you want. However, typical durations are:

SIP Holding Period Options

- **Minimum SIP Duration:** Some mutual funds allow minimum SIP tenure of 6 months or 1 year.
- **Long-Term SIP:**
 - 5+ years: Ideal for equity funds and wealth creation.
 - 10–15 years: Benefits from compounding and reduced risk.
 - 20+ years: Suitable for retirement or long-term goals.

3.6 SIP vs SWP

Feature	SIP (Systematic Investment Plan)	SWP (Systematic Withdrawal Plan)
Purpose	Invest fixed amounts regularly to build wealth over time	Withdraw fixed amounts regularly from investments to generate income
Suitable For	Long-term savings and rupee-cost averaging	Retirement income or recurring expenses
How It Works	Fixed amount invested at intervals in a mutual fund	Fixed amount withdrawn at intervals from mutual fund holdings
Example	Investing Rs 5,000 monthly	Withdrawing Rs 5,000 monthly

Table 3.1: Difference between SIP and SWP

3.7 SIP vs PPF

Feature	SIP (Systematic Investment Plan)	PPF (Public Provident Fund)
Investment Type	Market-linked mutual fund	Government-backed savings scheme
Risk	High (market risk)	Low (government guaranteed)
Returns	Market-dependent; potentially higher	Fixed and tax-free, but lower
Flexibility	High; can modify or stop anytime	Low; 15-year lock-in with partial withdrawal after 7 years
Taxation	Capital gains tax may apply	Fully tax-free
Lock-in Period	Usually no lock-in (except ELSS)	15 years

Table 3.2: Difference between SIP and PPF

Chapter 4

Data Analysis

Data Source:

The mutual fund data used for this analysis has been sourced from the Groww App and the official AMFI (Association of Mutual Funds in India) website. These platforms provide comprehensive and up-to-date information on various mutual fund schemes, including NAV, AUM, category, returns, and expense ratios.

About AMFI:

The **Association of Mutual Funds in India (AMFI)** is a non-profit industry body comprising all Asset Management Companies (AMCs) registered with the Securities and Exchange Board of India (SEBI). Incorporated on August 22, 1995 under Section 25 of the Companies Act, 1956 (now Section 8 of the Companies Act, 2013), AMFI aims to develop the mutual fund industry in a professional, ethical, and investor-friendly manner.

- AMFI operates under the guidance of a Board of fifteen elected Directors representing various AMCs.
- It functions cooperatively through standing committees such as:
 - Financial Literacy Committee
 - Operations Compliance Committee
 - Risk Committee
 - Certified Distributors (ARN) Committee
 - Equity CIOs Committee
 - Valuation Committee (Fixed Income CIOs)
 - ETF Committee
- AMFI is headquartered in Mumbai, Maharashtra, India.

- One of its flagship initiatives is the *Mutual Funds Sahi Hai* campaign, launched in 2017 under SEBI's guidance to increase investor awareness and promote mutual fund investments.
- Under SEBI regulations, all mutual fund distributors must register with AMFI and obtain an AMFI Registration Number (ARN).

Data Description

Prominent AMC's in India Offering SIP Options

- Aditya Birla Sun Life Mutual Fund
- Axis Mutual Fund
- Canara Robeco Mutual Fund
- DSP Mutual Fund
- Edelweiss Mutual Fund
- Franklin Templeton Mutual Fund
- HDFC Mutual Fund
- HSBC Mutual Fund
- ICICI Prudential Mutual Fund
- IDFC Mutual Fund
- Invesco Mutual Fund
- Kotak Mutual Fund
- LIC Mutual Fund
- Mirae Asset Mutual Fund
- Motilal Oswal Mutual Fund
- Nippon India Mutual Fund
- SBI Mutual Fund
- Sundaram Mutual Fund
- Tata Mutual Fund
- UTI Mutual Fund
- WhiteOak Capital Mutual Fund

Data

Fund Name	Category	AMC	AUM (in Cr)	Expense Ratio (%)
Aditya Birla Sun Life Medium Term Plan Direct Growth	Debt	Aditya Birla AMC	2144.49	0.86
Aditya Birla Sun Life Gold Fund Direct Growth	Commodities	Aditya Birla AMC	512.23	0.20
Aditya Birla Sun Life Balanced Advantage Fund Direct Growth	Hybrid	Aditya Birla AMC	6988.25	0.75
Aditya Birla Sun Life PSU Equity Fund Direct Growth	Equity	Aditya Birla AMC	4650.81	0.55
Axis Regular Saver Fund Direct Growth	Debt	Axis AMC	–	0.97
Axis Gold Direct Plan Growth	Commodities	Axis AMC	869.00	0.17
Axis Balanced Advantage Fund Direct Growth	Hybrid	Axis AMC	2625.19	0.83
Axis Small Cap Fund Direct Growth	Equity	Axis AMC	20954.45	0.57
Canara Robeco Income Direct Growth	Debt	Canara Robeco AMC	119.31	0.77
Canara Robeco Equity Hybrid Fund Direct Growth	Hybrid	Canara Robeco AMC	9841.01	0.64

Risk Level	NAV (Current)	NAV (1 Yr Ago)	NAV (3 Yrs Ago)	NAV (5 Yrs Ago)	SIP (Rs)	Total Invested (Rs)	CAGR (1 Yr)	CAGR (3 Yrs)	CAGR (5 Yrs)
Moderately High Risk	41.66	36.87	28.10	23.31	5000	300000	0.1299	0.1403	0.1231
High Risk	26.34	19.85	15.27	13.11	5000	300000	0.3270	0.1993	0.1497
Very High Risk	109.61	107.30	142.05	96.42	5000	300000	0.0215	-0.0828	0.0260
Very High Risk	30.69	36.16	28.23	11.72	5000	300000	-0.1513	0.0282	0.2123
Moderately High Risk	33.28	31.01	26.94	–	5000	300000	0.0732	0.0730	–
High Risk	27.90	21.20	17.10	14.30	5000	300000	0.3160	0.1773	0.1430
Very High Risk	21.68	19.85	14.67	11.82	5000	300000	0.0922	0.1391	0.1290
Very High Risk	105.10	98.85	63.05	–	5000	300000	0.0632	0.1857	–
Moderate Risk	60.16	55.80	53.50	50.00	5000	300000	0.0781	0.0399	0.0377
Very High Risk	364.99	348.26	253.56	181.70	5000	300000	0.0480	0.1291	0.1497

Final Value (Rs) One Yr	Final Value (Rs) Three Yr	Final Value (Rs) Five Yr	XIRR (One Yr) %	XIRR (Three Yr) %	XIRR (Five Yr) %
64394.50	224800.80	416031.45	12.91	13.20	12.85
71763.48	247713.93	448090.43	35.46	19.27	15.82
60704.30	158771.73	320675.88	2.04	-7.04	2.59
55303.90	188056.71	536138.25	-13.30	2.53	23.10
62433.11	201769.10	–	7.09	6.66	–
71331.60	238840.47	439704.55	34.11	16.97	15.06
63082.27	224358.66	422823.16	9.01	13.08	13.49
62095.11	242192.72	–	6.10	17.85	–
62601.05	191509.22	330597.49	7.59	3.59	3.78
61584.41	220764.89	448032.33	4.60	12.09	15.81

- **Asset Under Management (AUM)** : AUM refers to the market value of the total fund or the assets the fund holds.
- **Expense Ratio** : The annual fee expressed as a percentage of a fund's assets.
- **NAV(Net Asset Value)**: The per-unit value of a mutual fund scheme, calculated as the total value of assets minus liabilities, divided by the total number of outstanding units.
- **SIP Ammount**: The fixed amount invested regularly (typically monthly) by an investor in a mutual fund via a Systematic Investment Plan (SIP).
- **Total Investment**: The cumulative amount invested by the investor over time through SIPs or lump sum investments
- **CAGR**: The mean annual growth rate of an investment over a specified time period longer than one year, assuming the profits are reinvested annually.
- **Final Value**: The total value of the investment at the end of the investment period, including capital gains and reinvestments.
- **XIRR**: A method to calculate the annualized return on investments that are made at irregular intervals, commonly used for SIPs.
- **SIP Return**: The total returns generated from investments made via SIPs, typically measured using XIRR to account for periodic cash flows.

4.1 Table: SIP Returns of Top 70 Mutual Funds in India

Top 5 Funds of 1-Year Return Based on XIRR

Fund Name	Category	Risk	1 Year (%)
Invesco India Gold ETF FOF Direct Growth	Commodities	High Risk	37.95
Kotak Gold Fund Direct Growth	Commodities	High Risk	36.21
HDFC Gold ETF Fund of Fund Direct Plan Growth	Commodities	High Risk	35.95
Aditya Birla Sun Life Gold Fund Direct Growth	Commodities	High Risk	35.46
ICICI Prudential Regular Gold Savings (FOF) Direct Growth	Commodities	High Risk	34.35

Top 5 Funds of 3-Year Return Based on XIRR

Fund Name	Category	Risk	3 Year (%)
Motilal Oswal Midcap Fund Direct Growth	Equity	Very High Risk	28.39
DSP India T.I.G.E.R. (The Infrastructure Growth and Economic Reforms Fund) Direct Growth	Equity	Very High Risk	21.21
LIC MF Infrastructure Fund Direct Growth	Equity	Very High Risk	25.57
Edelweiss Mid Cap Direct Plan Growth	Equity	Very High Risk	25.44
Invesco India Small Cap Fund Direct Growth	Equity	Very High Risk	23.32

Top 5 Funds of 5-Year Return Based on XIRR

Fund Name	Category	Risk	5 Year (%)
Edelweiss Mid Cap Direct Plan Growth	Equity	Very High Risk	79.81
Motilal Oswal Midcap Fund Direct Growth	Equity	Very High Risk	33.61
DSP India T.I.G.E.R. (The Infrastructure Growth and Economic Reforms Fund) Direct Growth	Equity	Very High Risk	32.42
Invesco India Small Cap Fund Direct Growth	Equity	Very High Risk	30.67
Tata Digital India Fund Direct Growth	Equity	Very High Risk	30.56

Different Sector Wise Lump Sum and SIP values**Infrastructure Funds**

Fund Name	Category	CAGR(1 Yr)	CAGR(3 Yr)	CAGR(5 Yr)	XIRR(1 Yr)%	XIRR(3 Yr)%	XIRR(5 Yr)%
DSP India T.I.G.E.R. Direct Growth	Equity	2.57	27.21	28.76	2.44	27.21	32.42
ICICI Prudential Infrastructure Direct Growth	Equity	2.21	22.01	21.92	2.09	21.49	23.93
LIC MF Infrastructure Fund Direct Growth	Equity	8.40	25.75	24.33	8.18	25.57	26.85

Gold & Precious Metals Funds

Fund Name	Category	CAGR(1 Yr)	CAGR(3 Yr)	CAGR(5 Yr)	XIRR(1 Yr)%	XIRR(3 Yr)%	XIRR(5 Yr)%
Aditya Birla Sun Life Gold Fund Direct Growth	Commodities	32.70	19.93	14.97	35.46	19.27	15.85
Axis Gold Direct Plan Growth	Commodities	31.60	17.73	14.30	34.11	16.97	15.06
HDFC Gold ETF Fund of Fund Direct Plan Growth	Commodities	33.09	16.73	13.58	35.95	15.94	14.26
ICICI Prudential Regular Gold Savings FOF Direct Growth	Commodities	31.80	20.25	14.48	34.35	19.61	15.26
Invesco India Gold ETF FOF Direct Growth	Commodities	34.68	17.01	13.47	37.95	16.23	14.13
Kotak Gold Fund Direct Growth	Commodities	33.30	19.38	14.05	36.21	18.69	14.78

Debt & Hybrid Funds

Fund Name	Category	CAGR(1 Yr)	CAGR(3 Yr)	CAGR(5 Yr)	XIRR(1 Yr)%	XIRR(3 Yr)%	XIRR(5 Yr)%
Aditya Birla Sun Life Medium Term Plan Direct Growth	Debt	12.99	14.03	12.31	12.91	13.2	12.85
Franklin India Pension Direct Plan Growth	Hybrid	6.39	10.72	9.79	6.16	9.94	10.1
HDFC Credit Risk Debt Fund Direct Growth	Debt	8.19	6.91	7.54	7.97	6.3	7.7
HSBC Medium Duration Fund Direct Growth	Debt	8.43	7.05	6.73	8.21	6.43	6.85
ICICI Prudential All Seasons Bond Fund Direct Plan Growth	Debt	8.52	7.88	7.61	8.3	7.21	7.77
Invesco India Credit Risk Fund Direct Growth	Debt	6.37	7.52	7.13	6.14	6.87	7.26
Kotak Dynamic Bond Fund Direct Growth	Debt	7.84	7.09	7.00	7.61	6.47	7.13
LIC MF Medium to Long Duration Fund Direct Growth	Debt	9.09	6.95	5.96	8.87	6.33	6.03
Mahindra Manulife Low Duration Fund Direct Growth	Debt	7.91	6.29	5.16	7.68	5.71	5.21
Nippon India Ultra Short Duration Fund Direct Growth	Debt	7.70	7.04	6.99	7.47	6.41	7.11
SBI Magnum Gilt Fund Direct Growth	Debt	7.41	7.82	6.82	7.19	7.15	6.93
Sundaram Short Duration Fund Direct Growth	Debt	8.30	6.39	7.89	8.08	5.81	8.07
Tata Money Market Fund Direct Growth	Debt	7.63	7.13	5.95	7.41	6.5	6.02
UTI Dynamic Bond Fund Direct Growth	Debt	7.88	9.10	9.61	7.66	8.38	9.91

Descriptive Statistics

Table 4.1: Summary Statistics for CAGR and XIRR

	1 Year		3 Year		5 Year	
	Mean	SD	Mean	SD	Mean	SD
CAGR (%)	10.4	10.0	13.2	6.88	14.9	8.91
XIRR (%)	10.7	10.9	12.4	6.96	15.0	11.5

Principal Component Analysis

In mutual fund analysis, we often deal with multivariate data, such as multiple return metrics, risk indicators, or factor exposures. Visualizing such high-dimensional data using standard tools like scatterplots becomes impractical beyond three variables. To address this, we apply Principal Component Analysis (PCA). This enables us to capture the underlying structure of the data and represent it in 2 or 3 dimensions for easier interpretation and visualization.

Table 4.2: Importance of Principal Components

Component	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	1.9179	1.4976	0.8585	0.5797	0.06145	0.03709	0.02831
Proportion of Variance	0.5255	0.3204	0.1053	0.0480	0.00054	0.00020	0.00011
Cumulative Proportion	0.5255	0.8459	0.9512	0.9991	0.99969	0.99989	1.00000

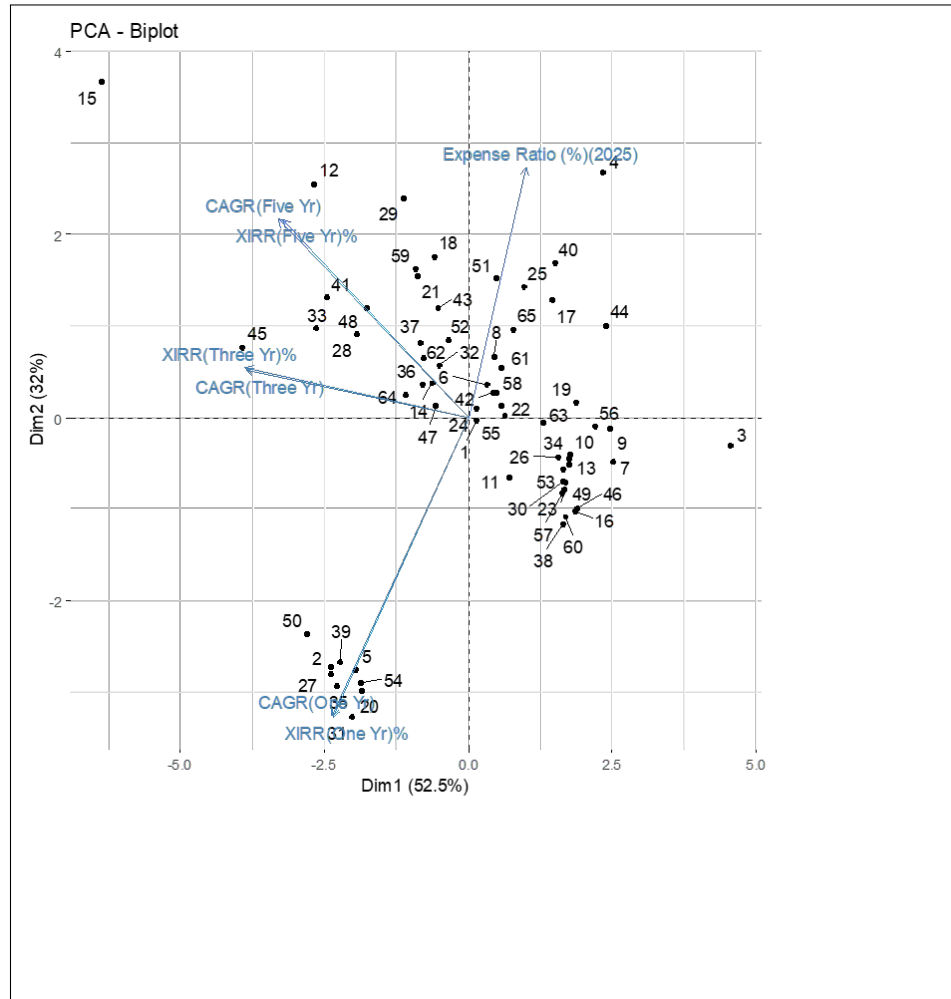


Figure 4.1: PCA

Interpretation: Dim1 and Dim2 are the first two principal components, together explaining approximately 84.5% of the total variance in the data.

Arrow Directions and Lengths:

- *Arrow Direction:* Indicates the direction of increasing values for each variable.

- *Arrow Length*: Longer arrows suggest a stronger influence on the principal components.

Examples:

- *CAGR(One Yr)* and *XIRR(One Yr)%* have long arrows pointing to the bottom-left, showing strong negative influence on both dimensions.
- *Expense Ratio (%) (2025)* points top-right, suggesting it influences observations in that direction.
- *CAGR(Five Yr)* and *XIRR(Five Yr)%* point in a different direction, indicating distinct return behavior.

Variable Relationships:

- Small angles between arrows indicate positive correlation.
- Arrows in opposite directions suggest negative correlation.
- Example: Expense Ratio appears negatively correlated with short-term return metrics (CAGR and XIRR One Yr).

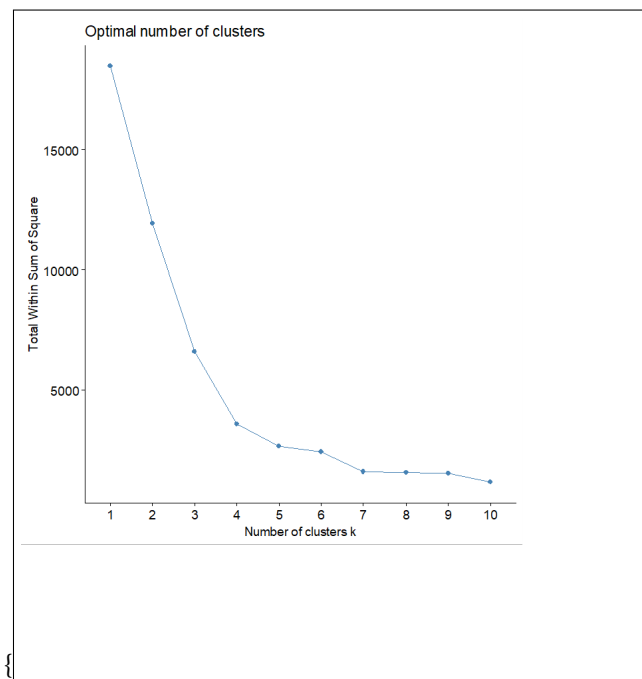


Figure 4.2: Elbow Method for Determining Optimal Clusters (K)

Point Clustering:

- Points close together represent schemes with similar performance profiles.

- Clusters in the lower-left or upper-right indicate funds with similar returns and expense ratios.

Interpretation: The Elbow Curve illustrates the within-cluster sum of squares (WSS) as the number of clusters k increases.

- There is a sharp decline in WSS from $k = 1$ to $k = 3$ or 4.
- After $k = 4$, the rate of decrease becomes significantly more gradual.
- This indicates that most of the variance is explained by 4 clusters, and adding more clusters beyond this point provides minimal improvement in compactness. Therefore, the optimal number of clusters is likely $k = 4$.

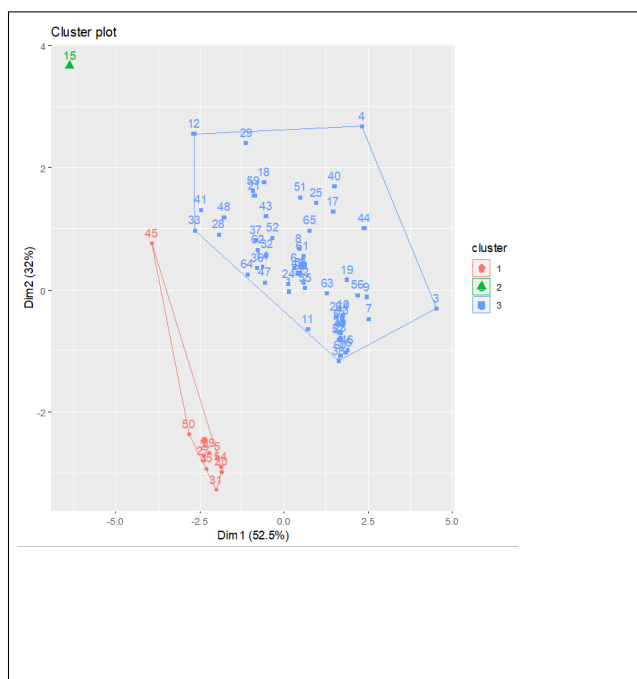


Figure 4.3: clustering

Table 4.3: Cluster-wise Interpretation and Investment Advice

Cluster	Description	Investment Advice
1 (Red)	Low performance, high cost	Avoid (X)
2 (Green)	Outlier (e.g., Fund 15)	Investigate Carefully (!)
3 (Blue)	Balanced / Strong performers	Consider Investing (✓)

4.2 Time Series Analysis

4.2.1 Invesco India Smallcap Fund - Direct Plan - Growth

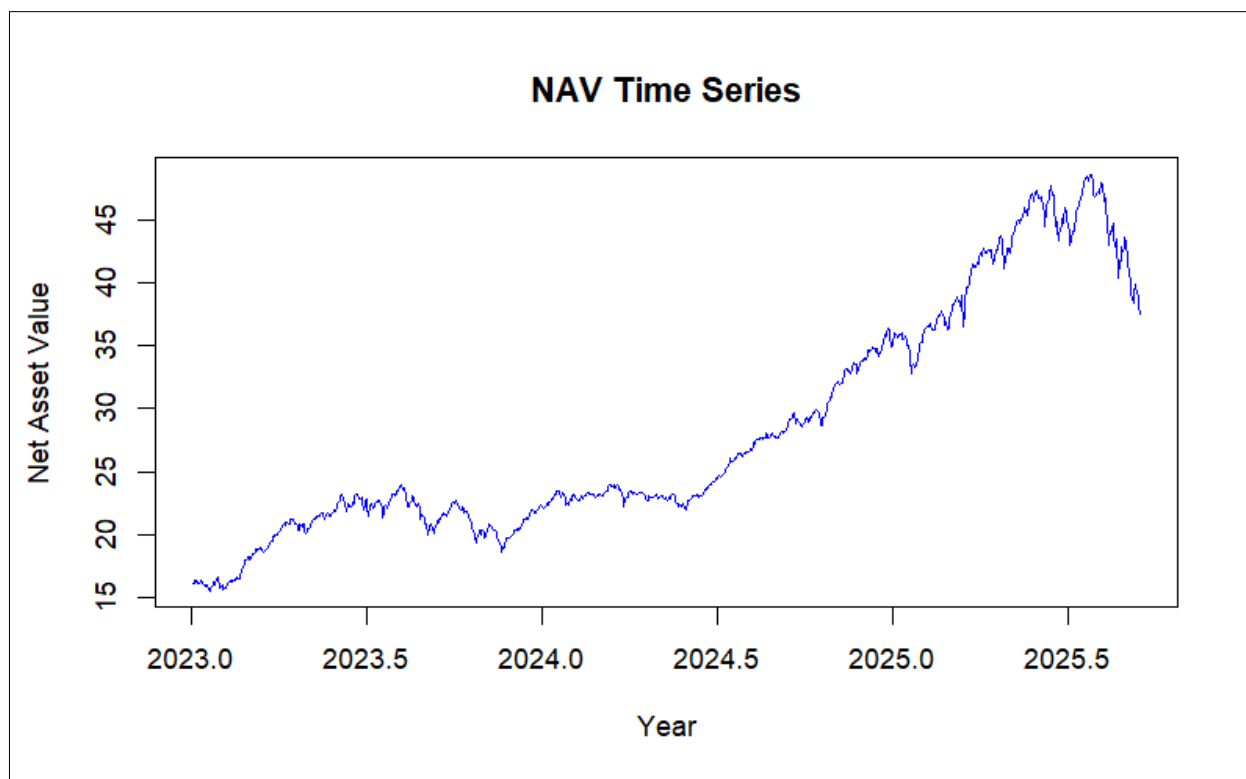


Figure 4.4: NAV TS

Interpretation:

- Upward Trend (2023 to early 2025):
- The NAV shows a steady and strong increase, indicating consistent growth in the fund's value.
- This suggests good performance, possibly driven by favorable market conditions or strong fund management.
- Peak and Decline (Mid-2025): The NAV peaks around mid-2025 and then declines sharply.
- This drop may signal market correction, volatility, or profit booking.
- Volatility in Later Phase: Towards the end, the NAV fluctuates significantly, indicating increased market uncertainty or risk exposure.

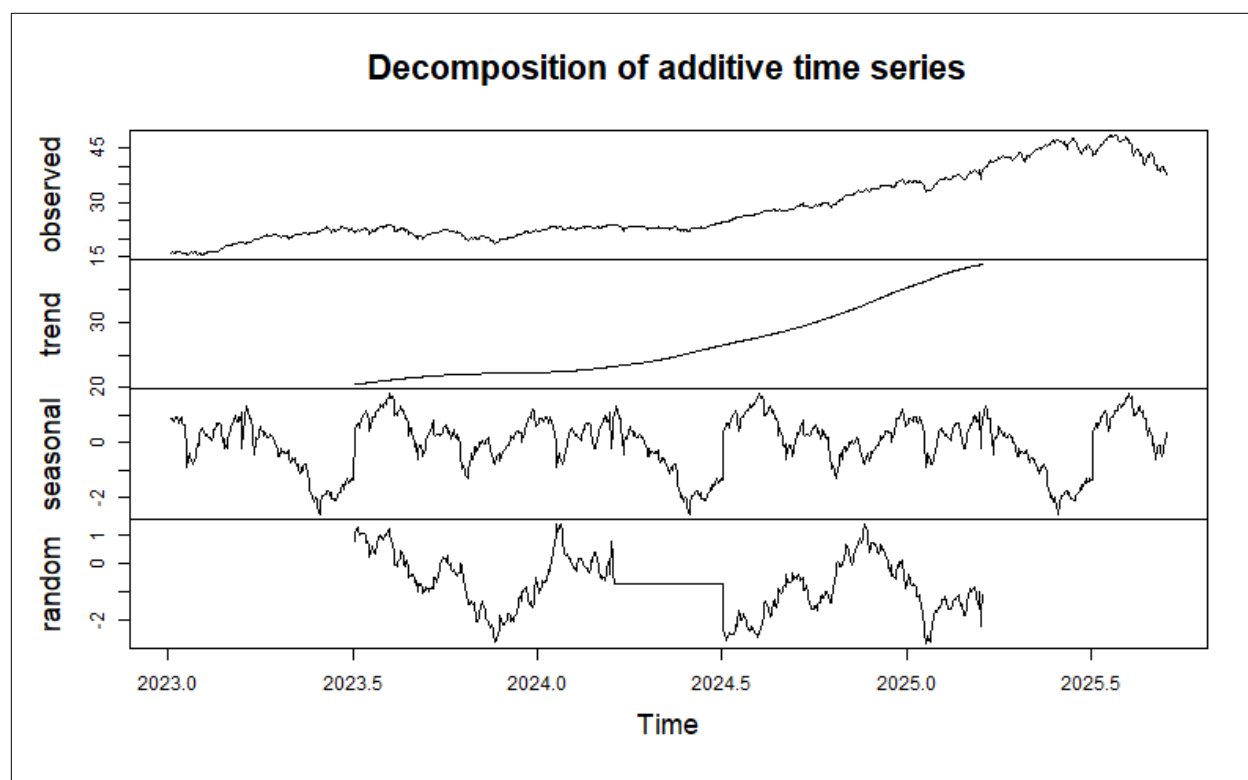


Figure 4.5

Interpretation

- **Observed (Top Panel)** This is the original NAV time series.
- It shows an upward trend from 2023 to mid-2025 with some fluctuations, peaking around mid-2025 and then slightly declining.
- **Trend (Second Panel)** The underlying long-term movement in the NAV. Indicates a steady and continuous upward trend, especially after mid-2024, implying consistent fund growth over time.
- **Seasonal (Third Panel)** Repeating seasonal patterns across years. Shows regular fluctuations around a central line, indicating that some months/periods tend to perform better or worse consistently (e.g., due to quarterly market behavior or economic cycles).
- **Random/Residual (Bottom Panel)** The irregular component—what remains after removing trend and seasonality. These are unexplained fluctuations, likely caused by short-term events or noise. Larger residuals indicate greater unpredictability in certain periods (e.g., early 2024 and early 2025).

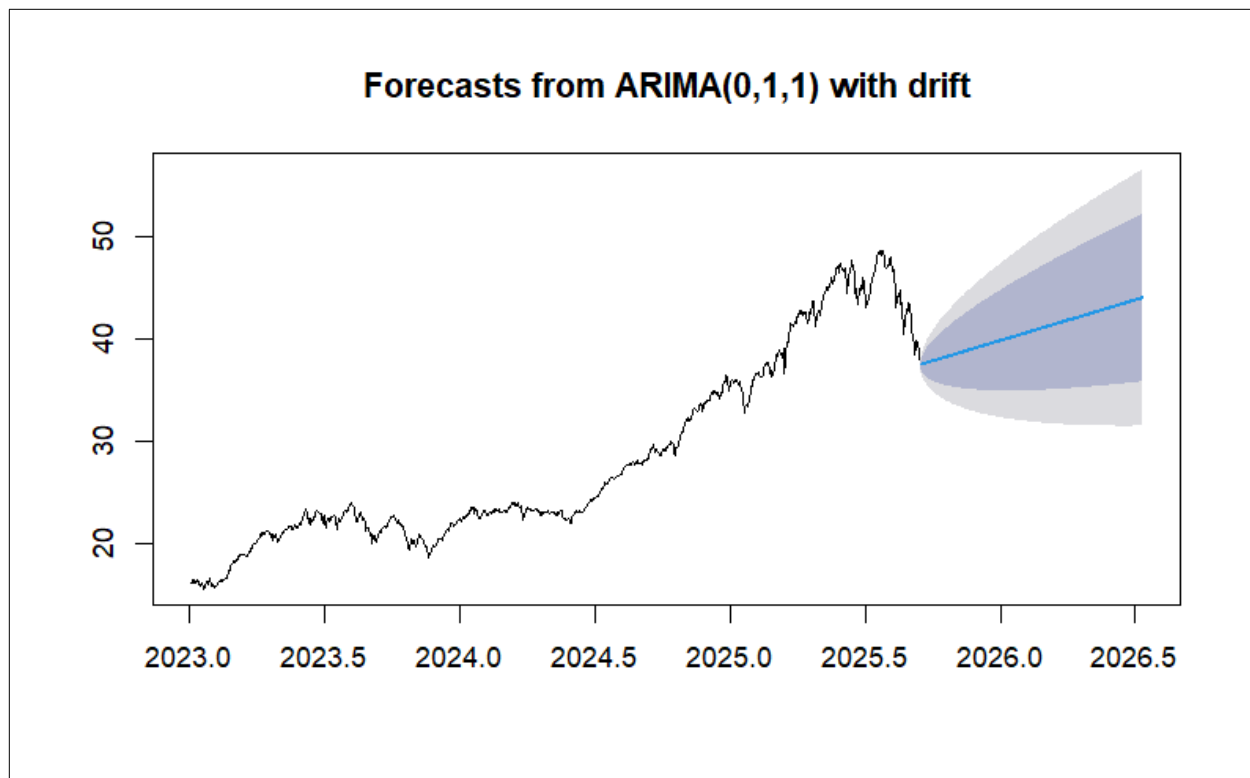
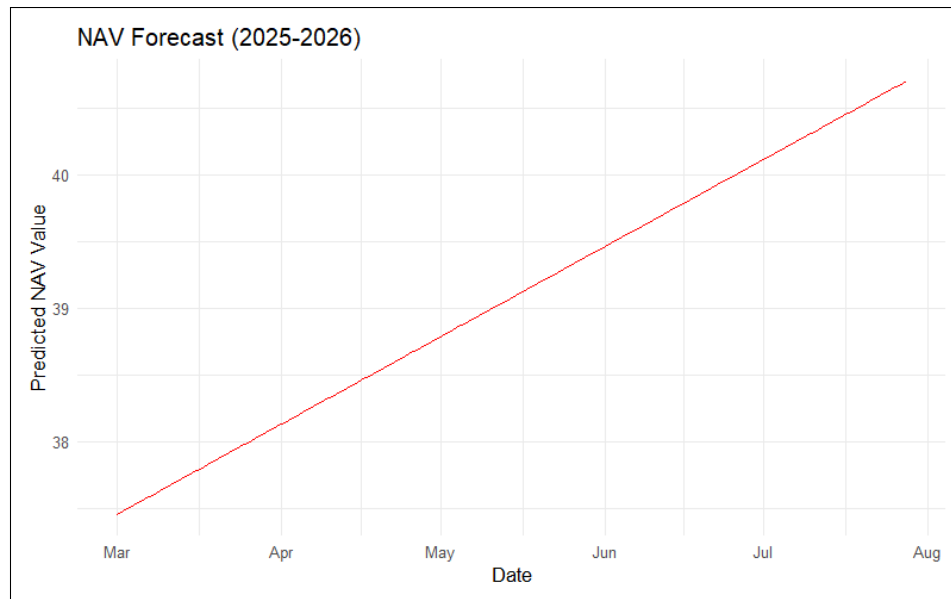


Figure 4.6: Forecasted NAV using ARIMA Model

Interpretation

- **Historical NAV (2023 to mid-2025):** The black line shows past NAV values, which rose steadily from Rs. 15 to a peak of over Rs. 45 by mid-2025.
- This period includes both strong upward trends and a recent decline, possibly due to market correction or volatility.
- **Forecast (mid-2025 to 2026.5):** The blue line represents the forecasted NAV trend based on the ARIMA model.
- It suggests a gradual upward movement (positive drift), meaning the model expects the NAV to recover and increase moderately over time.
- **Prediction Intervals (Shaded area):** The shaded region (dark to light blue) represents the 80% and 95% confidence intervals.
- It widens over time, indicating growing uncertainty the further you predict into the future.
- The NAV is most likely to stay within this band, but extremes outside the band are still possible.



Interpretation

- This forecast predicts a gradual and consistent increase in NAV from March to August 2025. While it offers a clear optimistic outlook, the lack of variability or confidence intervals means it may oversimplify the risk and volatility typically seen in mutual fund NAVs.

Predicted NAVs (Next 150 Days)

Date	Invesco India Smallcap Fund	Motilal Oswal Midcap Fund	Edelweiss Mid Cap Fund
09-05-2025	38.96132	105.8729	100.06242
10-05-2025	38.98309	105.9395	100.11627
11-05-2025	39.00487	106.0060	100.17012
12-05-2025	39.02665	106.0726	100.22397
13-05-2025	39.04843	106.1392	100.27782
14-05-2025	39.07021	106.2058	100.33167
15-05-2025	39.09199	106.2724	100.38552
16-05-2025	39.11377	106.3390	100.43936
17-05-2025	39.13555	106.4056	100.49321
18-05-2025	39.15732	106.4721	100.54706
19-05-2025	39.17910	106.5387	100.60091
20-05-2025	39.20088	106.6053	100.65476
21-05-2025	39.22266	106.6719	100.70861
22-05-2025	39.24444	106.7385	100.76246

Date	Invesco India Smallcap Fund	Motilal Oswal Midcap Fund	Edelweiss Mid Cap Fund
23-05-2025	39.26622	106.8051	100.81631
24-05-2025	39.28800	106.8716	100.87016
25-05-2025	39.30978	106.9382	100.92400
26-05-2025	39.33156	107.0048	100.97785
27-05-2025	39.35333	107.0714	101.03170
28-05-2025	39.37511	107.1380	101.08555
29-05-2025	39.39689	107.2046	101.13940
30-05-2025	39.41867	107.2712	101.19325
31-05-2025	39.44045	107.3377	101.24710
01-06-2025	39.46223	107.4043	101.30095
02-06-2025	39.48401	107.4709	101.35480
03-06-2025	39.50579	107.5375	101.40864
04-06-2025	39.52757	107.6041	101.46249
05-06-2025	39.54934	107.6707	101.51634
06-06-2025	39.57112	107.7373	101.57019
07-06-2025	39.59290	107.8038	101.62404
08-06-2025	39.61468	107.8704	101.67789
09-06-2025	39.63646	107.9370	101.73174
10-06-2025	39.65824	108.0036	101.78559
11-06-2025	39.68002	108.0702	101.83944
12-06-2025	39.70180	108.1368	101.89328
13-06-2025	39.72357	108.2034	101.94713
14-06-2025	39.74535	108.2699	102.00098
15-06-2025	39.76713	108.3365	102.05483
16-06-2025	39.78891	108.4031	102.10868
17-06-2025	39.81069	108.4697	102.16253
18-06-2025	39.83247	108.5363	102.21638
19-06-2025	39.85425	108.6029	102.27023
20-06-2025	39.87603	108.6695	102.32408
21-06-2025	39.89781	108.736	102.37792
22-06-2025	39.91958	108.8026	102.43177
23-06-2025	39.94136	108.8692	102.48562
24-06-2025	39.96314	108.9358	102.53947
25-06-2025	39.98492	109.0024	102.59332

Date	Invesco India Smallcap Fund	Motilal Oswal Midcap Fund	Edelweiss Mid Cap Fund
26-06-2025	40.0067	109.069	102.64717
27-06-2025	40.02848	109.1355	102.70102
28-06-2025	40.05026	109.2021	102.75487
29-06-2025	40.07204	109.2687	102.80872
30-06-2025	40.09381	109.3353	102.86256
01-07-2025	40.11559	109.4019	102.91641
02-07-2025	40.13737	109.4685	102.97026
03-07-2025	40.15915	109.5351	103.02411
04-07-2025	40.18093	109.6016	103.07796
05-07-2025	40.20271	109.6682	103.13181
06-07-2025	40.22449	109.7348	103.18566
07-07-2025	40.24627	109.8014	103.23951
08-07-2025	40.26805	109.868	103.29335
09-07-2025	40.28982	109.9346	103.3472
10-07-2025	40.3116	110.0012	103.40105
11-07-2025	40.33338	110.0677	103.4549
12-07-2025	40.35516	110.1343	103.50875
13-07-2025	40.37694	110.2009	103.5626
14-07-2025	40.39872	110.2675	103.61645
15-07-2025	40.4205	110.3341	103.6703
16-07-2025	40.44228	110.4007	103.72415
17-07-2025	40.46406	110.4673	103.77799
18-07-2025	40.48583	110.5338	103.83184
19-07-2025	40.50761	110.6004	103.88569
20-07-2025	40.52939	110.667	103.93954
21-07-2025	40.55117	110.7336	103.99339
22-07-2025	40.57295	110.8002	104.04724
23-07-2025	40.59473	110.8668	104.10109
24-07-2025	40.61651	110.9334	104.15494
25-07-2025	40.63829	110.9999	104.20879
26-07-2025	40.66006	111.0665	104.26263
27-07-2025	40.68184	111.1331	104.31648
28-07-2025	40.70362	111.1997	104.37033

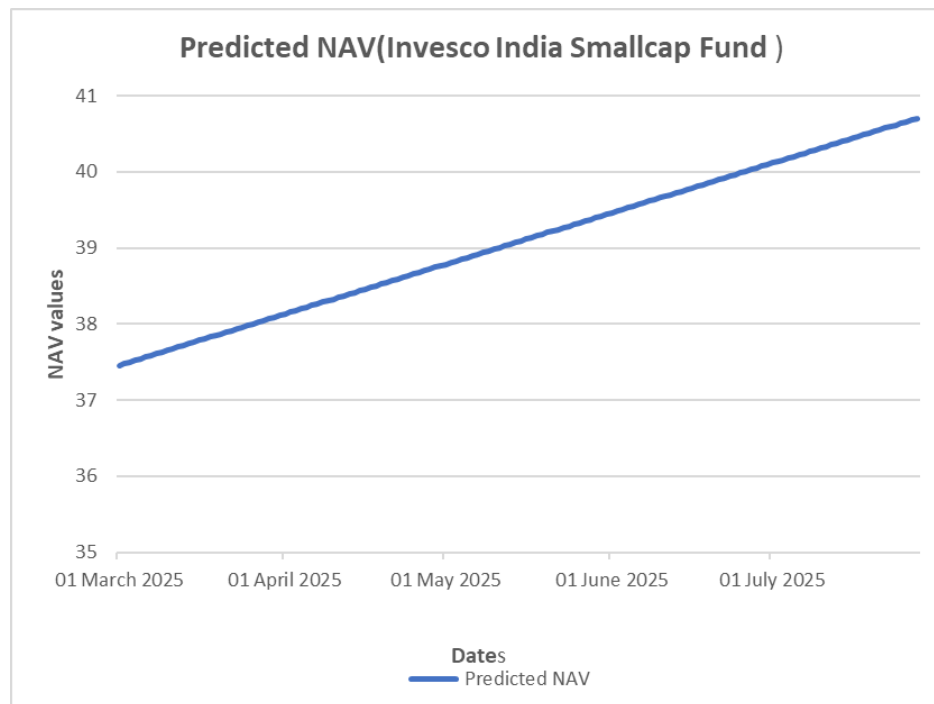


Figure 4.7

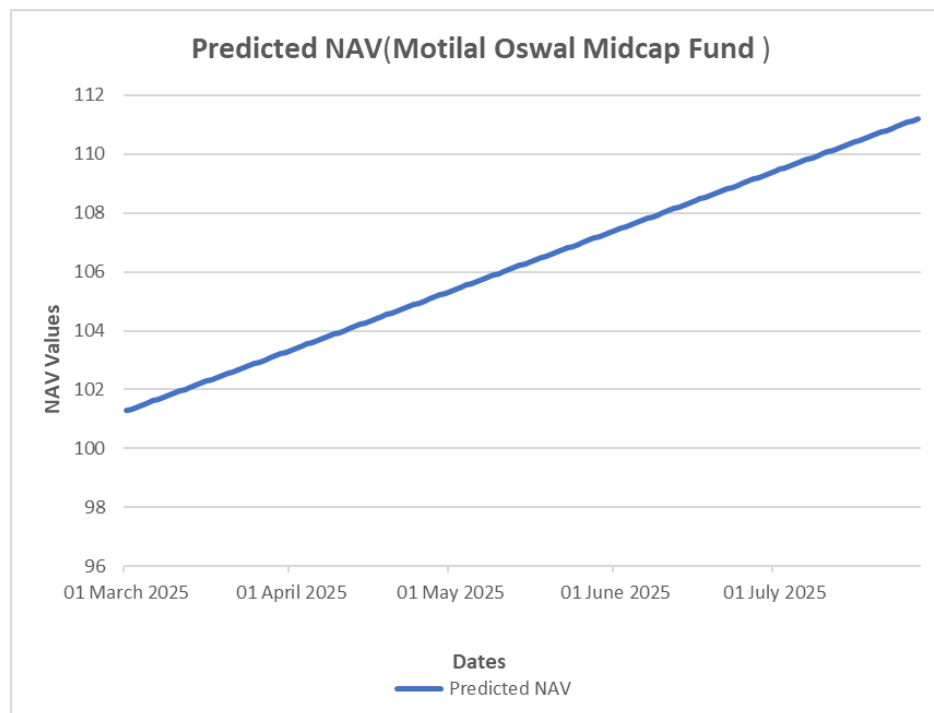


Figure 4.8

Comparison of SIP values and Lump sum values

TABLE SHOWING CALCULATION OF SIP AND LUMP SUM VALUE OF INVESCO INDIA SMALL CAP FUND DIRECT GROWTH								
		SIP Inverstment					Lump sum Inverstment	
NAV date1	NAV1	Inverstment	Units	Total Units	Total Inverstment	Value1	Total Unit	Value2
01-Mar-21	16.0652381	10000	622.461985	622.461985	10000	10000	29878.1753	480000
01-Apr-21	16.05368421	10000	622.909973	1245.371958	20000	19992.80814	29878.2	479655.1876
01-May-21	17.1315	10000	583.720048	1829.092006	30000	31335.0897	29878.2	511858.3833
01-Jun-21	18.77954545	10000	532.494251	2361.586257	40000	44349.51647	29878.2	561099.015
01-Jul-21	20.48047619	10000	488.269897	2849.856154	50000	58366.41112	29878.2	611919.7637
01-Aug-21	20.69095238	10000	483.303031	3333.159185	60000	68966.23798	29878.2	618208.4134
01-Sep-21	21.45714286	10000	466.045273	3799.204458	70000	81520.07281	29878.2	641100.8057
01-Oct-21	22.3625	10000	447.177194	4246.381652	80000	94959.7097	29878.2	668151.2475
01-Nov-21	22.59368421	10000	442.601565	4688.983218	90000	105941.4061	29878.2	675058.6156
01-Dec-21	22.36478261	10000	447.131554	5136.114772	100000	114868.0903	29878.2	668219.4477
01-Jan-22	23.174	10000	431.518081	5567.632852	110000	129024.3237	29878.2	692397.4068
01-Feb-22	21.699	10000	460.85073	6028.483583	120000	130812.0653	29878.2	648327.0618
01-Mar-22	21.18809524	10000	471.963142	6500.446725	130000	137732.0843	29878.2	633062.1471
01-Apr-22	22.16578947	10000	451.145673	6951.592397	140000	154087.5336	29878.2	662273.8911
01-May-22	20.22714286	10000	494.385197	7445.977594	150000	150610.8525	29878.2	604350.6197
01-Jun-22	19.77772727	10000	505.619269	7951.596863	160000	157264.5141	29878.2	590922.891
01-Jul-22	20.69095238	10000	483.303031	8434.899894	170000	174526.112	29878.2	618208.4134
01-Aug-22	22.1105	10000	452.273807	8887.1737	180000	196499.8541	29878.2	660621.9411
01-Sep-22	22.93545455	10000	436.006183	9323.179884	190000	213831.3684	29878.2	685270.098
01-Oct-22	22.96777778	10000	435.392579	9758.572463	200000	224132.7237	29878.2	686235.858
01-Nov-22	23.18904762	10000	431.238064	10189.81053	210000	236292.0015	29878.2	692847.0026
01-Dec-22	23.445	10000	426.530177	10616.3407	220000	248900.1078	29878.2	700494.399
01-Jan-23	23.17619048	10000	431.477296	11047.818	230000	256046.3343	29878.2	692462.8543
01-Feb-23	22.9685	10000	435.378888	11483.19689	240000	263751.8077	29878.2	686257.4367
01-Mar-23	22.6925	10000	440.674232	11923.87112	250000	270582.4454	29878.2	678011.0535
01-Apr-23	25.15857143	10000	397.478848	12321.34997	260000	309987.5633	29878.2	751692.8289
01-May-23	24.33136364	10000	410.992172	12732.34214	270000	309795.2466	29878.2	726977.349
01-Jun-23	26.06904762	10000	383.596675	13115.93882	280000	341920.0336	29878.2	778896.2186
01-Jul-23	27.29761905	10000	366.332316	13482.27113	290000	368033.9013	29878.2	815603.7214
01-Aug-23	27.97954545	10000	357.403948	13839.67508	300000	387227.818	29878.2	835978.455
01-Sep-23	29.00684211	10000	344.746249	14184.42133	310000	411445.2698	29878.2	866672.2298
01-Oct-23	29.3505	10000	340.709698	14525.13103	320000	426319.8582	29878.2	876940.1091
01-Nov-23	31.4305	10000	318.162295	14843.29332	330000	466532.1307	29878.2	939086.7651
01-Dec-23	33.326	10000	300.066015	15143.35934	340000	504667.5932	29878.2	995720.8932
01-Jan-24	34.55666667	10000	289.379763	15432.7391	350000	533304.0208	29878.2	1032490.998
01-Feb-24	35.75952381	10000	279.645782	15712.38488	360000	561867.4012	29878.2	1068430.204
01-Mar-24	34.41	10000	290.613194	16002.99807	370000	550663.1637	29878.2	1028108.862
01-Apr-24	36.6525	10000	272.832685	16275.83076	380000	596549.8869	29878.2	1095110.726
01-May-24	37.73142857	10000	265.031046	16540.86181	390000	624110.3457	29878.2	1127347.169
01-Jun-24	40.25736842	10000	248.401731	16789.26354	400000	675891.5677	29878.2	1202817.705
01-Jul-24	42.555	10000	234.990013	17024.25355	410000	724467.1098	29878.2	1271466.801
01-Aug-24	43.46	10000	230.096641	17254.35019	420000	749874.0593	29878.2	1298506.572
01-Sep-24	46.40238095	10000	215.506183	17469.85637	430000	810642.9306	29878.2	1386419.619
01-Oct-24	45.8285	10000	218.204829	17688.0612	440000	810617.3128	29878.2	1369273.089
01-Nov-24	44.8455	10000	222.987814	17911.04902	450000	803229.9486	29878.2	1339902.818
01-Dec-24	47.72619048	10000	209.528561	18120.57758	460000	864826.137	29878.2	1425972.664
01-Jan-25	44.32217391	10000	225.620702	18346.19828	470000	813143.3908	29878.2	1324266.777
01-Feb-25	40.44105263	10000	247.273484	18593.47176	480000	751939.5702	29878.2	1208305.859
TOTAL INVESTMENT AMOUNT						480000		480000
TOTAL UNIT						18593.47176		29878.2
AVERAGE UNIT PRICE						25.81551235		16.06522481

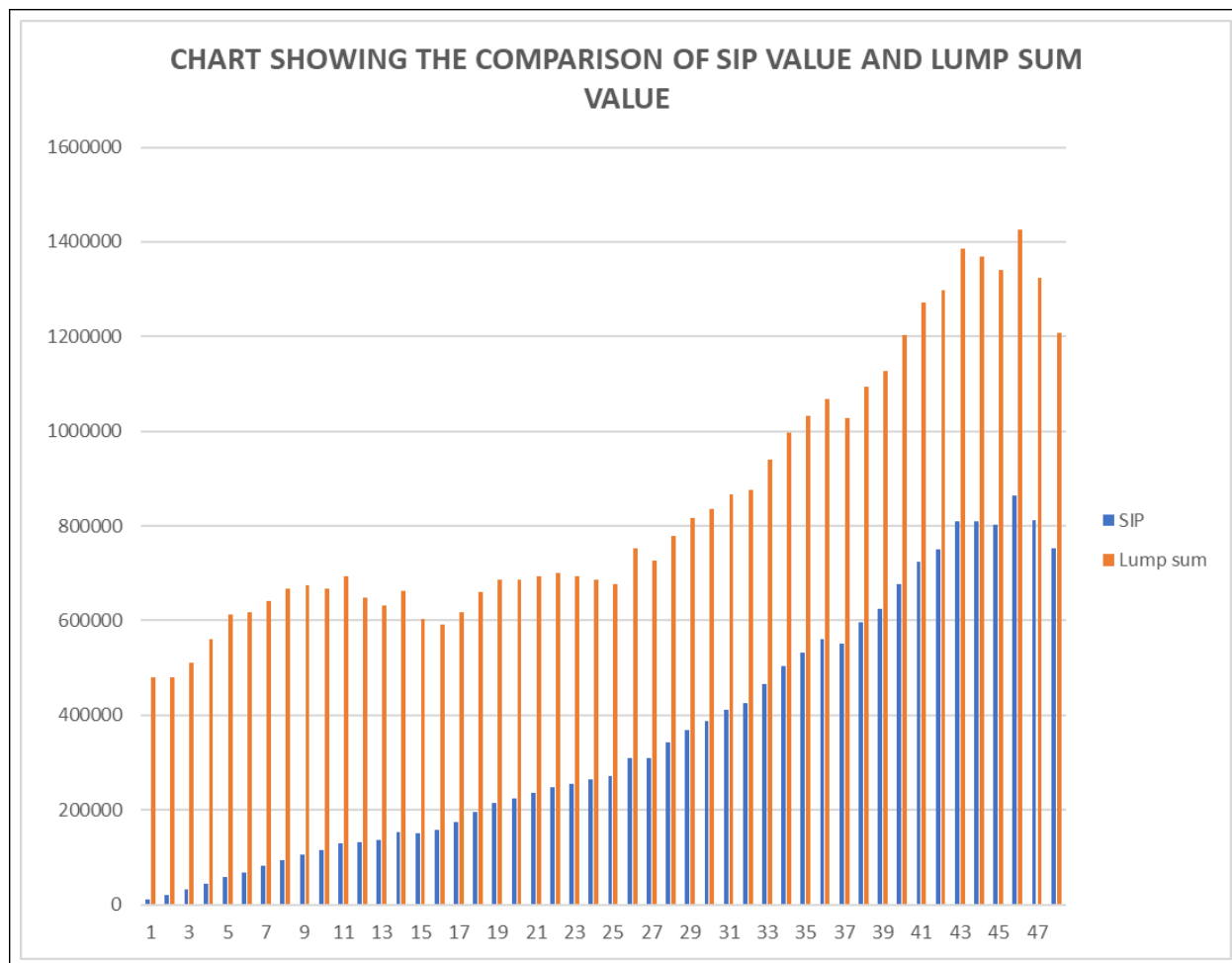


Figure 4.9: SIP vs Lump sum

interpretation

- Interpretation: This graph shows the comparison of SIP value and Lump Sum value of INVESCO INDIA SMALL CAP FUND DIRECT GROWTH. In March 21 the NAV of the mutual fund is 16.07. Then NAV is increased gradually in every year. So the unit purchased in SIP is decreased. After 5 years SIP value is increased up to 751939.5702. But in the case of LSIP the unit purchased is constant so its value increased with increase of NAV and its value is 1208305.859

4.3 Factor Exposure Analysis using Henriksson-Merton(HM)

Objective:

The purpose of this analysis is to examine how different categories of mutual funds (large cap, mid cap, small cap, and multi cap) respond to a specific risk factor — using the Henriksson-Merton (HM) regression model. This model allows us to test whether fund returns are significantly affected by a given factor, and in which direction (positive or negative).

Fund Selection:

We selected four mutual funds from each cap category (large, mid, small, multi), giving a total of 16 funds.

Model Used: The HM model is used as a time-series regression model that evaluates the market timing ability of fund managers by analyzing the exposure to factors like market risk, value, momentum, and size.

Table 4.5: Henriksson-Merton Model Results for Mutual Funds

Fund	Category	Alpha (HM)	Gamma (HM)	p-value (HM)	R-squared (HM)
ABSLMF	Large Cap	-0.00105	0.00278	0.00019	0.48118
MOMF	Large Cap	0.00125	-0.00043	0.47166	0.85741
DSP	Large Cap	0.00044	-0.00045	0.08398	0.89100
LIC	Large Cap	0.00028	-0.00047	0.00907	0.95429
ABSLMF	Small Cap	0.00150	-0.00193	0.00299	0.53028
MOMF	Small Cap	0.00293	-0.00299	0.02675	0.49552
DSP	Small Cap	0.00142	-0.00136	0.04021	0.46559
LIC	Small Cap	0.00158	-0.00118	0.37937	0.38304
ABSLMF	Mid Cap	0.00119	-0.00134	0.01199	0.64990
MOMF	Mid Cap	0.00202	-0.00197	0.00266	0.53223
DSP	Mid Cap	0.00096	-0.00117	0.02307	0.63623
LIC	Mid Cap	0.00132	-0.00103	0.26780	0.60006
ABSLMF	Multi Cap	0.00109	-0.00141	0.00011	0.80621
MOMF	Multi Cap	0.00284	-0.00100	0.64129	0.42116
DSP	Multi Cap	0.00154	-0.00176	0.08511	0.61685
LIC	Multi Cap	0.00114	-0.00093	0.12627	0.63236

Henriksson-Merton Model Results: Market Return Timing Analysis

This section presents the results from the Henriksson-Merton (HM) model, which analyzes the market return timing ability of fund managers. The model evaluates the magnitude (Alpha) and direction (Gamma) of the fund managers' timing ability, along with the statistical significance (p-value) and the explanatory power (R-squared).

4.3.1 Large-Cap Funds

- **ABSLMF:** The fund shows a slight underperformance with an Alpha of -0.00105. The Gamma is positive at 0.00278, suggesting a small but positive market timing ability. The p-value of 0.00019 indicates statistical significance, and the R-squared of 0.48118 reflects moderate explanatory power.
- **MOMF:** This fund shows positive performance with an Alpha of 0.00125. However, the Gamma is very small (-0.00043), indicating minimal market timing ability. The p-value of 0.47166 suggests the result is not statistically significant, although the R-squared of 0.85741 indicates strong explanatory power.
- **DSP:** The Alpha is slightly positive (0.00044), and the Gamma is negative (-0.00045), suggesting a slightly negative market timing ability. The p-value (0.08398) is marginally significant, and the R-squared is quite high at 0.891.
- **LIC:** LIC shows minimal outperformance with an Alpha of 0.00028. The negative Gamma (-0.00047) suggests weak market timing ability. The p-value of 0.00907 is significant, and the R-squared of 0.95429 is very high.

4.3.2 Small-Cap Funds

- **ABSLMF:** This fund shows positive performance with an Alpha of 0.00150, but the Gamma is negative (-0.00193), indicating weak market timing ability. The p-value (0.00299) is significant, and the R-squared (0.53028) shows moderate explanatory power.
- **MOMF:** MOMF demonstrates positive performance with an Alpha of 0.00293, but the negative Gamma (-0.00299) indicates poor market timing ability. The p-value of 0.02675 suggests statistical significance, and the R-squared is moderate at 0.49552.
- **DSP:** The fund's Alpha is 0.00142, suggesting positive performance. The Gamma is negative (-0.00136), indicating weak market timing. The p-value (0.04021) is significant, and the R-squared (0.46559) is moderate.
- **LIC:** LIC has a positive Alpha of 0.00158, but the negative Gamma (-0.00118) indicates weak market timing ability. The p-value of 0.37937 is not statistically significant, and the R-squared (0.38304) is relatively low.

4.3.3 Mid-Cap Funds

- **ABSLMF:** This fund has a positive Alpha of 0.00119, but the negative Gamma (-0.00134) suggests weak market timing ability. The p-value (0.01199) is statistically significant, and the R-squared (0.64990) shows moderate explanatory power.
- **MOMF:** MOMF shows a positive Alpha of 0.00202 but a negative Gamma (-0.00197), indicating weak market timing ability. The p-value (0.00266) is significant, and the R-squared (0.53223) shows moderate explanatory power.
- **DSP:** The fund's Alpha (0.00096) is positive, but the negative Gamma (-0.00117) indicates weak market timing. The p-value (0.02307) is significant, and the R-squared (0.63623) is moderate.
- **LIC:** LIC has a positive Alpha of 0.00132, but the negative Gamma (-0.00103) suggests weak market timing. The p-value (0.26780) is not statistically significant, and the R-squared (0.60006) is moderate.

4.3.4 Multi-Cap Funds

- **ABSLMF:** ABSLMF shows a positive Alpha of 0.00109, but the negative Gamma (-0.00141) suggests weak market timing ability. The p-value (0.00011) is statistically significant, and the R-squared (0.80621) reflects high explanatory power.
- **MOMF:** MOMF demonstrates positive performance with an Alpha of 0.00284, but the negative Gamma (-0.00100) indicates weak market timing ability. The p-value (0.64129) is not significant, and the R-squared (0.42116) is low.
- **DSP:** The Alpha is 0.00154, indicating positive performance, but the Gamma (-0.00176) suggests weak market timing ability. The p-value (0.08511) is marginally significant, and the R-squared (0.61685) is moderate.
- **LIC:** LIC shows a positive Alpha (0.00114) with weak market timing ability (Gamma of -0.00093). The p-value (0.12627) is not statistically significant, and the R-squared (0.63236) is moderate.

Conclusion: From the results, it is evident that most of the funds across categories exhibit positive Alphas, indicating some level of outperformance. However, the market timing ability (Gamma) for most funds is weak or negative, particularly in the small-cap and mid-cap categories. Large-cap and multi-cap funds show more promising market timing results, but they are still limited in terms of magnitude and direction. These findings suggest that, while fund managers may achieve outperformance, their ability to time the market effectively is constrained, especially in smaller fund categories.

4.3.5 SIP contribution remain consistent

Table 4.6: SIP Contribution by Financial Year (in Rs. Crore)

Month	FY 2024-25	FY 2023-24	FY 2022-23	FY 2021-22	FY 2020-21	FY 2019-20	FY 2018-19	FY 2017-18	FY 2016-17
Total during FY	2,89,352	1,99,219	1,55,972	1,24,566	96,080	1,00,084	92,693	67,190	43,921
March	25,926	19,271	14,276	12,328	9,182	8,641	8,055	7,119	4,335
February	25,999	19,187	13,686	11,438	7,528	8,513	8,095	6,425	4,050
January	26,400	18,838	13,856	11,517	8,023	8,532	8,064	6,644	4,095
December	26,459	17,610	13,573	11,305	8,418	8,518	8,022	6,222	3,973
November	25,320	17,073	13,306	11,005	7,302	8,273	7,985	5,893	3,884
October	25,323	16,928	13,041	10,519	7,800	8,246	7,985	5,621	3,434
September	24,509	16,042	12,976	10,351	7,788	8,263	7,727	5,516	3,698
August	23,547	15,814	12,693	9,923	7,792	8,231	7,658	5,206	3,497
July	23,332	15,245	12,140	9,609	7,831	8,324	7,554	4,947	3,334
June	21,262	14,734	12,276	9,156	7,917	8,122	7,554	4,744	3,310
May	20,904	14,749	12,286	8,819	8,123	8,183	7,304	4,584	3,189
April	20,371	13,728	11,863	8,596	8,376	8,238	6,690	4,269	3,122

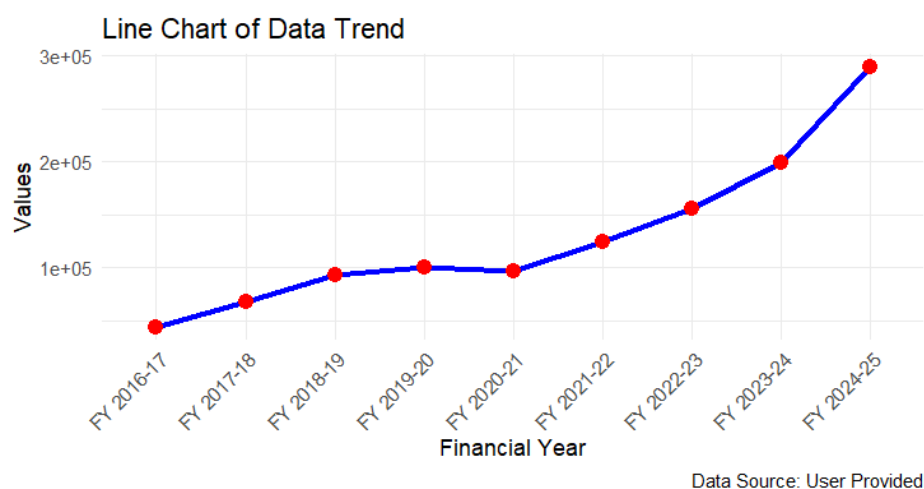


Figure 4.10: SIP contribution

The Power of Compounding

Compounding is a powerful financial concept where the returns on an investment are reinvested to generate additional earnings over time. The key idea is that the longer you invest, the more significant the effect of compounding becomes. In a Systematic Investment Plan (SIP), regular investments grow at a compounded rate, resulting in exponential growth over the years. Even small amounts invested consistently can lead to significant wealth accumulation when compounded over long periods. To demonstrate the power of compounding, let's consider

the example of a SIP, where a person invests a fixed amount every month. The returns generated by the investments themselves start earning interest, accelerating the growth of the overall amount.

Key Points about Compounding:

- The longer you invest, the more your returns grow exponentially.
- Small increases in the rate of return can lead to significant growth over time.
- Starting early can result in a larger corpus compared to investing later in life, even if the monthly contribution is the same.

Let's now look at the visual representation of how compounding works in a SIP over time using the Shiny app we developed.

Shiny App: SIP Calculator

Below are screenshots from the Shiny app showing how different input values (monthly investment, years, and ROI) affect the final future value and total investment. These demonstrate the power of compounding over different time periods.

Simple SIP Calculator

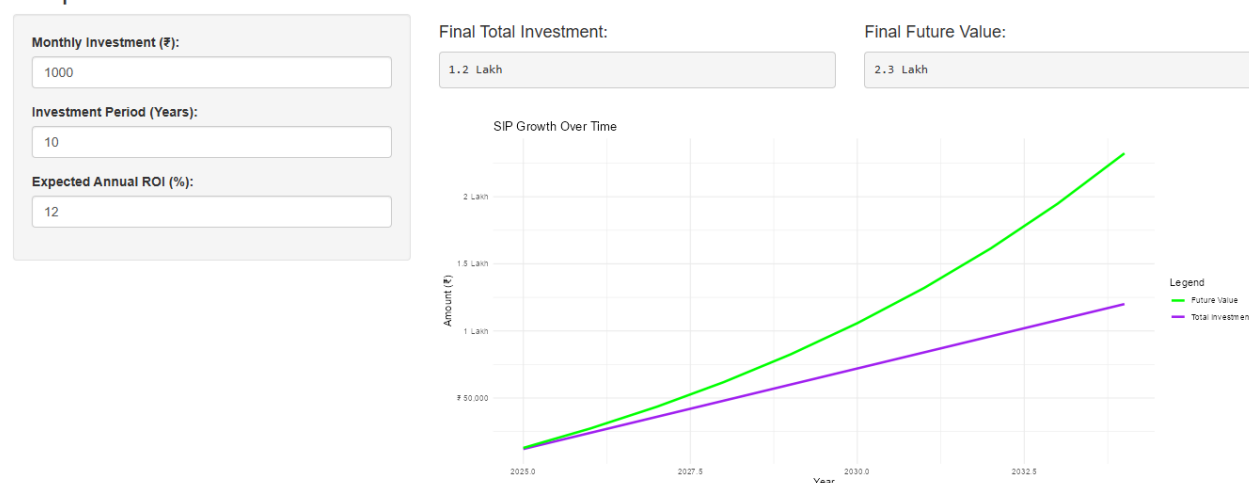


Figure 4.11: Shiny app

Simple SIP Calculator



Figure 4.12: shiny app

Chapter 5

Conclusion

Power of Compounding: SIPs demonstrate significant wealth-building potential over long durations due to the effect of compounding. This is well-illustrated using projections and Shiny app simulations for varying investment amounts, durations, and rates of return.

Comparative Analysis: SIPs, when compared to government schemes like **PPF**, **SSY**, and **NPS**, offer potentially higher returns, albeit with greater market-linked risks. Government schemes offer safety and tax benefits but are less flexible and have return caps.

Mutual Fund Categories: Detailed analysis of SIPs across different mutual fund categories—Equity (small-cap, mid-cap, large-cap, multi-cap)—was conducted using performance indicators like *Sharpe Ratio*, *Jensen's Alpha*, *Treynor Ratio*, etc., showcasing which funds outperformed based on risk-adjusted returns.

Fund Manager Timing Ability: The application of *Henriksson-Merton* and *Treynor-Mazuy* models effectively evaluated the market timing skills of fund managers. The results showed variability in timing ability across categories, with multi-cap and large-cap funds showing relatively better timing ability on average.

Recommendations:

- Long-term investors with a higher risk appetite should consider equity-based SIPs, especially diversified or multi-cap funds.
- Conservative investors may balance SIPs with safer options like PPF/NPS.
- Regular performance review and alignment with risk profile are crucial.

Dashboard Utility: The R Shiny dashboards integrated into this project provide an excellent interactive tool for visualizing returns, comparing schemes, and making informed decisions based on customized inputs.

Chapter 6

Appendix

```
1 import numpy as np
2 from scipy.optimize import newton
3 from datetime import datetime
4
5 # Define cash flows (Negative for SIPs, Positive for Redemption)
6 cashflows = [-5000] * 12 # Rs 5,000 SIP each month
7 cashflows.append(45893) # Redemption value on maturity
8
9 # Define corresponding dates
10 dates = [
11     "2024-03-06", "2024-04-06", "2024-05-06", "2024-06-06",
12     "2024-07-06", "2024-08-06", "2024-09-06", "2024-10-06",
13     "2024-11-06", "2024-12-06", "2025-01-06", "2025-02-06",
14     "2025-03-20" # Redemption Date
15 ]
16
17 # Convert dates to datetime format
18 dates = [datetime.strptime(date, "%Y-%m-%d") for date in dates]
19
20 # Convert dates to days since first investment
21 days = [(date - dates[0]).days for date in dates]
22
23 # Define the XIRR function (NPV = 0)
24 def xirr(rate):
25     return sum(cf / (1 + rate) ** (day / 365) for cf, day in zip(cashflows,
26                                                                    days))
27
28 # Solve for XIRR using Newton-Raphson method
29 xirr_value = newton(xirr, 0.1) # Initial guess of 10%
```

```
29
30 # Convert to percentage
31 xirr_percentage = round(xirr_value * 100, 2)
32
33 # Print the result
34 print(f"XIRR (Annualized Return): {xirr_percentage}%")
35
36
37
38 [caption={Python Code to Calculate 3-Year XIRR for SIP Investment}, label={lst
    :xirr-three-year}]
39 import numpy as np
40 from scipy.optimize import newton
41 from datetime import datetime, timedelta
42
43 # Define SIP investment details
44 sip_amount = -5000 # Rs 5,000 per month (Negative because it's an outflow)
45 months = 36 # 3 years of SIP (36 months)
46
47 redemption_value = 219007.5466 # Assumed maturity value after 3 years
48
49 # Generate SIP investment dates (6th of every month for 3 years)
50 start_date = datetime(2022, 3, 6) # Example start date (March 6, 2022)
51 dates = [start_date.replace(year=start_date.year + (i // 12), month=(i % 12) +
    1, day=6) for i in range(months)]
52 dates.append(datetime(2025, 3, 20)) # Redemption Date
53
54 # Define cash flows (SIPs are negative, final value is positive)
55 cashflows = [sip_amount] * months # SIP payments
56 cashflows.append(redemption_value) # Final value after 3 years
57
58 # Convert dates to days since first investment
59 days = [(date - dates[0]).days for date in dates]
60
61 # Define the XIRR function (NPV = 0)
62 def xirr(rate):
63     return sum(cf / (1 + rate) ** (day / 365) for cf, day in zip(cashflows,
        days))
64
65 # Solve for XIRR using Newton-Raphson method
66 xirr_value = newton(xirr, 0.1) # Initial guess of 10%
67
68 # Convert to percentage
```

```

69 xirr_percentage = round(xirr_value * 100, 2)
70
71 # Print the result
72 print(f"3-Year XIRR (Annualized Return): {xirr_percentage}%")
73
74
75
76
77 [caption={Python Code to Calculate 5-Year XIRR for SIP Investment}, label={lst
       :xirr-five-year}]
78 import numpy as np
79 from scipy.optimize import newton
80 from datetime import datetime
81 from dateutil.relativedelta import relativedelta # Corrects date shifting
82
83 # Define SIP investment details
84 monthly_sip = 5000 # Rs 5,000 per month
85 num_months = 60 # 5 years (60 months)
86
87 # Generate SIP cash flows (negative values for investment)
88 cashflows = [-monthly_sip] * num_months
89
90 # Assume redemption value at the end of 5 years
91 redemption_value = 473658.025 # Modify based on actual NAV
92 cashflows.append(redemption_value)_

```

Listing 6.1: Python Code to Calculate One-Year XIRR for SIP Investment

```

1 # Load required libraries
2 rm(list = ls())
3 library(quantmod)
4 library(writexl)
5
6 data <- read.csv("C:/ST-524/RDK/ABSL_mul.csv", stringsAsFactors = FALSE)
7
8 # Step 2: Check and convert the date
9 head(data$NAV.date)
10 data$NAV.date <- as.Date(data$NAV.date, format = "%d-%b-%y")
11 summary(data$NAV.date)
12 range(data$NAV.date)
13
14 # Step 3: Calculate fund return
15 data <- data[order(data$NAV.date), ]
16 data$Fund_Return <- c(NA, diff(log(data$Net.Asset.Value)))

```

```

17
18 # Step 4: Get Nifty 50 market return
19 getSymbols("^NSEI", src = "yahoo", from = "2019-01-01", to = "2024-12-31")
20 nifty_returns <- dailyReturn(Cl(NSEI), type = "log")
21 nifty_df <- data.frame(Date = index(nifty_returns), Market_Return = coredata(
    nifty_returns))
22 nifty_df$Date <- as.Date(nifty_df$Date)
23
24 # Step 5: Filter mutual fund data to match market dates
25 data <- data[data$NAV.date >= min(nifty_df$Date) & data$NAV.date <= max(nifty_
    df$Date), ]
26
27 # Step 6: Merge data
28 merged_data <- merge(data, nifty_df, by.x = "NAV.date", by.y = "Date", all =
    FALSE)
29 merged_data$Indicator <- ifelse(merged_data$daily.returns > 0, 1, 0)
30
31 # HenrikssonMerton model
32 hm_model <- lm(Fund_Return ~ daily.returns + Indicator, data = merged_data)
33 summary(hm_model)
34
35 # TreynorMazuy model
36 merged_data$Market_Square <- merged_data$daily.returns^2
37 tm_model <- lm(Fund_Return ~ daily.returns + Market_Square, data = merged_data
    )
38 summary(tm_model)
39
40 library(readxl)
41 library(dplyr)
42 library(stats)
43 # Load the dataset
44 df<- read_excel("D:/Msc-2 Sem-4 Reserach Project/Data/SIP_ (2).xlsx")
45 View(df)
46
47 # Convert XIRR(One Yr)% to numeric
48 df$`XIRR(One Yr)%` <- as.numeric(df$`XIRR(One Yr)%`)
49 # Extract top 5 highest `XIRR(One Yr)%` values
50 top_5 <- df %>%
51   arrange(desc(`XIRR(One Yr)%`), `SIP Return`) %>%
52   select(`Fund Name`, Category, `Risk Level`, `XIRR(One Yr)%`, `SIP Return`) %>%
53   head(10)
54 # Print the result
55 print(top_5)

```



```

56 # Convert XIRR(Three Yr)% to numeric
57 df$`XIRR(Three Yr)%` <- as.numeric(df$`XIRR(Three Yr)%`)
58 # Extract top 5 highest `XIRR(One Yr)%` values
59 top_5 <- df %>%
60   arrange(desc(`XIRR(Three Yr)%`)) %>%
61   select(`Fund Name`, Category, `Risk Level`, `XIRR(Three Yr)%`) %>%
62   head(10)
63
64 # Print the result
65 print(top_5)
66 ```
67
68
69 # Convert XIRR(Five Yr)% to numeric
70 df$`XIRR(Five Yr)%` <- as.numeric(df$`XIRR(Five Yr)%`)
71 # Extract top 5 highest `XIRR(Five Yr)%` values
72 top_5 <- df %>%
73   arrange(desc(`XIRR(Five Yr)%`)) %>%
74   select(`Fund Name`, Category, `Risk Level`, `XIRR(Five Yr)%`) %>%
75   head(10)
76 # Print the result
77 print(top_5)
78
79 # Load necessary libraries
80 rm(list=ls())
81 library(readxl)
82 library(dplyr)
83 library(ggplot2)
84 library(corrplot)
85 library(cluster)
86 library(factoextra)
87 library(tidyr)
88 library(caret)
89
90 # Load the dataset
91 df <- readxl::read_excel("D:/Msc-2 Sem-4 Reserach Project/Data/SIP_ (2).xlsx")
92 View(df)
93
94 # --- 1. Descriptive Statistics ---
95 summary_stats <- df %>%
96   summarise(across(c(`CAGR(One Yr)`, `CAGR(Three Yr)`, `CAGR(Five Yr)`, `XIRR(
97     One Yr)%`, `XIRR(Three Yr)%`, `XIRR(Five Yr)%`),
98     list(mean = mean, sd = sd), na.rm = TRUE))

```

```

98 print(summary_stats)
99
100 # --- 2. Correlation Analysis ---
101 cor_data <- df %>%
102   select(`CAGR(One Yr)`, `CAGR(Three Yr)`, `CAGR(Five Yr)`, `XIRR(One Yr)%`, `
103     XIRR(Three Yr)%`, `XIRR(Five Yr)%`, `Expense Ratio (%) (2025)` ) %>%
104   na.omit()
105
106 cor_matrix <- cor(cor_data)
107 corplot(cor_matrix, method = "circle")
108
109 # --- 3. PCA ---
110 pca_data <- cor_data
111 pca_result <- prcomp(pca_data, scale. = TRUE)
112 summary(pca_result)
113
114 # Visualize PCA
115 fviz_pca_biplot(pca_result, repel = TRUE)
116
117 # --- 4. Clustering (K-Means) ---
118 # Choose optimal number of clusters
119 fviz_nbclust(pca_data, kmeans, method = "wss")
120
121 # Apply K-means
122 set.seed(123)
123 kmeans_result <- kmeans(scale(pca_data), centers = 3)
124 df$Cluster <- as.factor(kmeans_result$cluster)
125
126 # Visualize Clusters
127 fviz_cluster(kmeans_result, data = scale(pca_data))
128
129 ```
130
131 ```{r}
132 #Historical NAV for a period From 01-Mar-2021 to 01-Mar-2025
133
134 #Invesco India Smallcap Fund - Direct Plan - Growth
135
136 rm(list=ls())
137 # Load necessary libraries
138 library(readxl)

```

```

139 library(ggplot2)
140 library(forecast)
141
142 data1<- read_excel("D:/Msc-2 Sem-4 Reserach Project/Code/Invesco india.xlsx")
143 #View(data)
144
145 # Convert NAV date to Date format
146 data1$`NAV date` <- as.Date(data1$`NAV date`, format="%m/%d/%Y")
147
148 # Create a time series object
149 nav_ts <- ts(data1$`Net Asset Value`, start = c(2023, 3), frequency = 365)
150
151 # Plot the time series
152 plot(nav_ts, main="NAV Time Series", xlab="Year", ylab="Net Asset Value", col=
  "blue")
153
154 # Decompose the time series
155 nav_decomp <- decompose(nav_ts)
156 plot(nav_decomp)
157
158 # Apply ARIMA model for forecasting
159 nav_arima <- auto.arima(nav_ts)
160 nav_forecast <- forecast(nav_arima, h=300)
161
162 # Plot the forecast
163 plot(nav_forecast)
164 # Print forecasted values
165 print(nav_forecast)
166
167 # Create a time series object
168 start_year <- year(min(data1$`NAV date`)) # Extract start year dynamically
169 start_month <- month(min(data1$`NAV date`))
170 nav_ts <- ts(data1$`Net Asset Value`, start = c(start_year, start_month),
  frequency = 365)
171
172 # Plot the historical time series
173 plot(nav_ts, main="Historical NAV Time Series", xlab="Year", ylab="Net Asset
  Value", col="blue")
174
175 # Apply ARIMA model for forecasting
176 nav_arima <- auto.arima(nav_ts)
177 nav_forecast <- forecast(nav_arima, h=150) # Predict for the next 5 Month
  (150days)

```

```

178
179 # Generate proper date sequence for forecasts
180 forecast_dates <- seq(from = max(data1$`NAV date`) + 1, length.out = 150, by =
    "days")
181
182 # Create a dataframe for easier reading
183 forecast_df1 <- data.frame(Date = forecast_dates, Predicted_NAV = nav_forecast
    $mean)
184
185 # Plot the extended forecast with proper dates
186 ggplot(forecast_df1, aes(x = Date, y = Predicted_NAV)) +
187   geom_line(color="red") +
188   ggtitle("NAV Forecast (2025-2026)") +
189   xlab("Date") +
190   ylab("Predicted NAV Value") +
191   theme_minimal()
192
193 # Print forecasted values with clear dates
194 print(forecast_df1)
195
196
197 ```
198
199
200 ```{r}
201 #Historical NAV for a period From 01-Mar-2021 to 01-Mar-2025
202
203 #Motilal Oswal Midcap Fund-Direct Plan-Growth Option
204 rm(list=ls())
205 # Load necessary libraries
206 library(readxl)
207 library(ggplot2)
208 library(forecast)
209
210 data2<- read_excel("D:/Msc-2 Sem-4 Reserach Project/Code/Motilal Oswal.xlsx")
211 #View(data)
212
213 # Convert NAV date to Date format
214 data2$`NAV date` <- as.Date(data2$`NAV date`, format="%m/%d/%Y")
215
216 # Create a time series object
217 nav_ts <- ts(data2$`Net Asset Value`, start = c(2023, 3), frequency = 365)

```

```

218 # Plot the time series
219 plot(nav_ts, main="NAV Time Series", xlab="Year", ylab="Net Asset Value", col=
    "blue")
220
221 # Decompose the time series
222 nav_decomp <- decompose(nav_ts)
223 plot(nav_decomp)
224
225 # Apply ARIMA model for forecasting
226 nav_arima <- auto.arima(nav_ts)
227 nav_forecast <- forecast(nav_arima, h=300)
228
229 # Plot the forecast
230 plot(nav_forecast)
231 # Print forecasted values
232 print(nav_forecast)
233 # Create a time series object
234 start_year <- year(min(data2$`NAV date`)) # Extract start year dynamically
235 start_month <- month(min(data2$`NAV date`))
236 nav_ts <- ts(data2$`Net Asset Value`, start = c(start_year, start_month),
    frequency = 365)
237
238 # Plot the historical time series
239 plot(nav_ts, main="Historical NAV Time Series", xlab="Year", ylab="Net Asset
    Value", col="blue")
240
241 # Apply ARIMA model for forecasting
242 nav_arima <- auto.arima(nav_ts)
243 nav_forecast <- forecast(nav_arima, h=150) # Predict for the next 5 Month
    (150days)
244
245 # Generate proper date sequence for forecasts
246 forecast_dates <- seq(from = max(data2$`NAV date`) + 1, length.out = 150, by =
    "days")
247
248 # Create a dataframe for easier reading
249 forecast_df2 <- data.frame(Date = forecast_dates, Predicted_NAV = nav_forecast
    $mean)
250
251 # Plot the extended forecast with proper dates
252 ggplot(forecast_df2, aes(x = Date, y = Predicted_NAV)) +
253   geom_line(color="red") +
254   ggtitle("NAV Forecast (2025-2026)") +

```

```
255   xlab("Date") +
256   ylab("Predicted NAV Value") +
257   theme_minimal()
258
259 # Print forecasted values with clear dates
260 print(forecast_df2)
261
262 ```
263
264
265
266 ```{r}
267 #Historical NAV for a period From 01-Mar-2021 to 01-Mar-2025
268
269 #Tata Digital India Fund
270 #Tata Digital India Fund-Direct Plan-Growth
271 rm(list=ls())
272 # Load necessary libraries
273 library(readxl)
274 library(ggplot2)
275 library(forecast)
276
277 data3<- read_excel("D:/Msc-2 Sem-4 Reserach Project/Code/Tata digital.xlsx")
278 #View(data)
279
280 # Convert NAV date to Date format
281 data3$`NAV date` <- as.Date(data3$`NAV date`, format="%m/%d/%Y")
282
283 # Create a time series object
284 nav_ts <- ts(data3$`Net Asset Value`, start = c(2023, 3), frequency = 365)
285
286 # Plot the time series
287 plot(nav_ts, main="NAV Time Series", xlab="Year", ylab="Net Asset Value", col=
  "blue")
288
289 # Decompose the time series
290 nav_decomp <- decompose(nav_ts)
291 plot(nav_decomp)
292
293 # Apply ARIMA model for forecasting
294 nav_arima <- auto.arima(nav_ts)
295 nav_forecast <- forecast(nav_arima, h=300)
```

```

296 # Plot the forecast
297 plot(nav_forecast)
298 # Print forecasted values
299 print(nav_forecast)
300 # Create a time series object
301 start_year <- year(min(data3$`NAV date`)) # Extract start year dynamically
302 start_month <- month(min(data3$`NAV date`))
303 nav_ts <- ts(data3$`Net Asset Value`, start = c(start_year, start_month),
304             frequency = 365)
305
306 # Plot the historical time series
307 plot(nav_ts, main="Historical NAV Time Series", xlab="Year", ylab="Net Asset
308       Value", col="blue")
309
310 # Apply ARIMA model for forecasting
311 nav_arima <- auto.arima(nav_ts)
312 nav_forecast <- forecast(nav_arima, h=150) # Predict for the next 5 Month
313       (150days)
314
315 # Generate proper date sequence for forecasts
316 forecast_dates <- seq(from = max(data3$`NAV date`) + 1, length.out = 150, by =
317       "days")
318
319 # Create a dataframe for easier reading
320 forecast_df3 <- data.frame(Date = forecast_dates, Predicted_NAV = nav_forecast
321       $mean)
322
323 # Plot the extended forecast with proper dates
324 ggplot(forecast_df3, aes(x = Date, y = Predicted_NAV)) +
325   geom_line(color="red") +
326   ggtitle("NAV Forecast (2025-2026)") +
327   xlab("Date") +
328   ylab("Predicted NAV Value") +
329   theme_minimal()
330
331 # Print forecasted values with clear dates
332 print(forecast_df3)
333
334 #Historical NAV for a period From 01-Mar-2021 to 01-Mar-2025
335
336 #Edelweiss Mid Cap Fund
337 #Edelweiss Mid Cap Fund - Direct Plan - Growth Option

```

```

333
334 rm(list=ls())
335 # Load necessary libraries
336 library(readxl)
337 library(ggplot2)
338 library(forecast)
339
340 data4<- read_excel("D:/Msc-2 Sem-4 Reserach Project/Code/Edelwiss mid cap.xlsx
    ")
341 #View(data)
342
343 # Convert NAV date to Date format
344 data4$`NAV date` <- as.Date(data4$`NAV date`, format="%m/%d/%Y")
345
346 # Create a time series object
347 nav_ts <- ts(data4$`Net Asset Value`, start = c(2023, 3), frequency = 365)
348
349 # Plot the time series
350 plot(nav_ts, main="NAV Time Series", xlab="Year", ylab="Net Asset Value", col=
    "blue")
351
352 # Decompose the time series
353 nav_decomp <- decompose(nav_ts)
354 plot(nav_decomp)
355
356 # Apply ARIMA model for forecasting
357 nav_arima <- auto.arima(nav_ts)
358 nav_forecast <- forecast(nav_arima, h=300)
359
360 # Plot the forecast
361 plot(nav_forecast)
362 # Print forecasted values
363 print(nav_forecast)
364 # Create a time series object
365 start_year <- year(min(data4$`NAV date`)) # Extract start year dynamically
366 start_month <- month(min(data4$`NAV date`))
367 nav_ts <- ts(data4$`Net Asset Value`, start = c(start_year, start_month),
    frequency = 365)
368
369 # Plot the historical time series
370 plot(nav_ts, main="Historical NAV Time Series", xlab="Year", ylab="Net Asset
    Value", col="blue")
371

```



```

372 # Apply ARIMA model for forecasting
373 nav_arima <- auto.arima(nav_ts)
374 nav_forecast <- forecast(nav_arima, h=150) # Predict for the next 5 Month
      (150days)
375
376 # Generate proper date sequence for forecasts
377 forecast_dates <- seq(from = max(data4$`NAV date`) + 1, length.out = 150, by =
      "days")
378
379 # Create a dataframe for easier reading
380 forecast_df4 <- data.frame(Date = forecast_dates, Predicted_NAV = nav_forecast
      $mean)
381
382 # Plot the extended forecast with proper dates
383 ggplot(forecast_df4, aes(x = Date, y = Predicted_NAV)) +
384   geom_line(color="red") +
385   ggtitle("NAV Forecast (2025-2026)") +
386   xlab("Date") +
387   ylab("Predicted NAV Value") +
388   theme_minimal()
389
390 # Print forecasted values with clear dates
391 print(forecast_df4)

```

Listing 6.2: R code for Market Timing Analysis

Chapter 7

References

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